Faithful Legal Text Simplification for Sri Lankan Statutes: A Hallucination-Controlled Framework using NLI Verification

Ashok Ainkaran Jeyathasan

CB013207

BSc (Hons) Computer Science

Supervised By: Mr. Krishnamoorthy Caucidheesan

Staffordshire University

June 15th 2026

DECLARATION

I declare that the research included in this senior project is wholly original. Every source used in the study has been properly cited and acknowledged. Additionally, this work has never before been presented for a degree or certification to any other university or institution.

Name: Ashok Ainkaran Jeyathasan

CB Number: CB013207

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my supervisor Mr. Krishnamoorthy Caucidheesan for his guidance, support and feedback throughout the development of this Final Year Project. His advice helped me refine the research direction, improve the technical approach and maintain focus on the main objective of developing a faithful legal text simplification system for Sri Lankan statutes.

I am also grateful to the academic staff of APIIT and Staffordshire University for the knowledge, guidance and learning environment provided throughout my degree programme. The technical and research skills gained during the programme were valuable in completing this project.

ABSTRACT

Legal statutes are drafted for precision rather than readability, which makes them difficult for non-lawyers to understand. At the same time, text generation systems can produce fluent rewrites that add unsupported or altered content, which is especially risky in legal contexts. This project addresses the need for safer public-facing legal simplification by focusing on Sri Lankan statute provisions and explicitly checking whether simplified statements are supported by the original text.

A prototype-driven web system is developed that accepts a statute provision as input and generates a plain-language simplification using prompt-based generation. The output is decomposed into atomic claims and verified using Natural Language Inference (NLI) where evidence is extracted from the same statute provision. The system produces a status-aware simplified output with supported/unsupported highlighting and evidence display to improve transparency.

The project will evaluate readability change using FKGL and ARI, semantic similarity using ROUGE-L and BERTScore, and faithfulness using claim-supportedness rates derived from NLI labels supported by a small manual review subset to check preservation of obligations, conditions, exceptions and definitions.

ACM Subject Descriptors (CCS):

- Computing methodologies → Natural language processing
- Applied computing → Law
- Human-centered computing → Web-based interaction

Keywords: Legal NLP; Legal text simplification; Sri Lankan statutes; hallucination control; factuality; natural language inference; claim extraction; legal AI.

Table of Contents

List of Figures

List of Tables

ABBREVIATIONS

AI — Artificial Intelligence

API — Application Programming Interface

ARI — Automated Readability Index

BERT — Bidirectional Encoder Representations from Transformers

BERTScore — BERT-based Semantic Similarity Score

CCS — ACM Computing Classification System

CSV — Comma-Separated Values

CSS — Cascading Style Sheets

DeBERTa — Decoding-enhanced BERT with Disentangled Attention

FKGL — Flesch-Kincaid Grade Level

FYP — Final Year Project

HTML — HyperText Markup Language

JSON — JavaScript Object Notation

LLM — Large Language Model

NLI — Natural Language Inference

NLP — Natural Language Processing

PDF — Portable Document Format

QA — Question Answering

RAG — Retrieval-Augmented Generation

ROUGE-L — Recall-Oriented Understudy for Gisting Evaluation, Longest Common Subsequence

SMART — Specific, Measurable, Achievable, Relevant, Time-bound

UI — User Interface

CHAPTER 01: INTRODUCTION

1.1 Chapter overview

This chapter introduces the research domain of legal NLP and legal text simplification, then defines the specific problem addressed by this project which is simplifying Sri Lankan statute text while controlling hallucinations through claim-level verification. It next explains the motivation and the research gap identified in existing work, before presenting the project's intended contributions and anticipated challenges. Finally, it states the research questions, the overall aim, and the SMART research objectives.

1.2 Problem background

This section provides background from three angles: (i) the broader domain context (legal NLP and LLM use), (ii) the key technological subdomain (legal text simplification and faithfulness), and (iii) the Sri Lankan practical context motivating the work.

1.2.1 Context of the research domain

Natural Language Processing (NLP) is increasingly used in the legal domain to support both professionals and laypersons, including tasks such as document analysis, information retrieval, and decision support (Ariai, Mackenzie and Demartini, 2025). However, legal language has domain-specific complexity and limited open datasets compared with many mainstream NLP areas, which creates barriers to building reliable legal systems across jurisdictions (Ariai, Mackenzie and Demartini, 2025).

Large Language Models (LLMs) have also attracted attention for legal applications such as drafting and legal research, but surveys emphasise practical risks including accuracy limitations and ethical concerns in real deployments (Dehghani et al., 2025). The need for reliability is particularly important in legal settings where outputs may influence understanding, decisions, and downstream actions (Magesh et al., 2025).

1.2.2 Subdomain or key technological context

A core problem in the legal domain is that legal documents are hard to understand for non-experts due to specialized terminology and complex syntax (Garimella et al., 2022; Cemri, Çukur and Koç, 2022). Legal text simplification aims to rewrite complex legal text in a more readable form while preserving meaning (Garimella et al., 2022; Cemri, Çukur and Koç, 2022).

At the same time, modern text generation systems can produce outputs that are fluent but not faithful to the source. In summarisation research, hallucinations are described as content that is not supported by the input document, and this is a known challenge for neural generation systems (Maynez et al., 2020). Surveys of LLM hallucination further describe the phenomenon as generating plausible but nonfactual or unverifiable content, and distinguish different hallucination types (Huang et al., 2025). These concerns motivate "hallucination-controlled" approaches where generation is paired with verification.

NLI provides a structured way to verify whether a statement is supported by evidence: given a *premise* and a *hypothesis*, the relationship is classified as entailment, contradiction, or neutral (Koreeda and Manning, 2021). NLI-based inconsistency detection has been explored in summarisation evaluation, including approaches that score factual consistency using NLI models (Laban et al., 2022; Scirè et al., 2024). This supports the feasibility of using claim-level verification for controlling unsupported outputs in a statute simplification workflow.

1.2.3 Practical / real-world context

Sri Lanka possesses a pluralistic legal system where many citizens struggle to understand information regarding fundamental rights, the Penal Code, or the Maintenance Act No. 37 of 1999 (Jayawardena et al., 2024). Surveys reveal that 55.4% of Sri Lankans face challenges due to linguistically inaccessible documents, and 89% view legal information access as a critical necessity (Fernando et al., 2024). This information asymmetry creates a significant barrier to justice for pro se litigants (Jayawardena et al., 2024).

Existing Sri Lankan-focused systems in the provided literature include legal question answering over Supreme Court case law embeddings (Jayawardena et al., 2024) and a constitution-focused chatbot motivated by limited legal awareness and lack of accessible legal context (Fernando et al., 2024).

However, question answering is not the same as rewriting statute provisions into simpler language, and the practical need remains for tools that support safer comprehension of statute text for non-lawyers while reducing the risk of incorrect paraphrases. This is aligned with broader legal NLP findings that legal datasets and tools are often jurisdiction-dependent and do not automatically generalise across legal systems (Akter et al., 2025).

1.3 Problem definition

This project addresses the problem of producing simplified versions of Sri Lankan statute sections without introducing unsupported content. Legal simplification is needed because legal language can be difficult for ordinary people to understand (Garimella et al., 2022; Cemri, Çukur and Koç, 2022). Yet generation-based systems can hallucinate or introduce inaccuracies, which is a known issue in text generation and summarisation (Maynez et al., 2020; Huang et al., 2025).

In legal contexts, empirical studies show that LLM-based legal answering can still produce fabricated or unsupported legal content, highlighting the need for explicit verification even when outputs appear confident (Dahl et al., 2024). This illustrates why a simplification system must include controls that explicitly check outputs against the source statute text rather than assuming the rewrite is faithful.

1.3.1 Problem statement

Sri Lankan citizens lack access to understandable legal statutes, and existing AI tools present a high risk of hallucinating unsupported legal advice, necessitating a domain-specific, NLI-verified simplification framework to ensure factual integrity.

1.4 Research motivation

The motivation is primarily societal: bridging the "access to justice" gap by empowering citizens with understandable legal information (Fernando et al., 2024). Technically, it responds to the failure of general models to distinguish between legal "obligations" and "permissions," which can result in imitative falsehoods (Dahl et al., 2024; Muralidharan, 2021).

1.5 Research gap

- **Gap 1: Limited Sri Lanka–specific focus on statute simplification workflows.** The Sri Lankan-focused systems provided in the literature emphasise question answering over case law and constitution retrieval rather than statute simplification (Jayawardena et al., 2024; Fernando et al., 2024). More broadly, legal NLP surveys highlight jurisdiction dependence and limited open datasets, which means approaches and tools often do not directly transfer across legal systems (Ariai, Mackenzie and Demartini, 2025; Akter et al., 2025).
- **Gap 2: Simplification work does not consistently integrate claim-level faithfulness controls into the user-facing output.** Legal simplification is motivated by comprehension barriers (Garimella et al., 2022; Cemri, Çukur and Koç, 2022), yet generation models are known to introduce unsupported content (Maynez et al., 2020; Huang et al., 2025). While NLI-based inconsistency detection is explored for summarisation evaluation (Laban et al., 2022; Scirè et al., 2024), this is not typically presented as an integrated "status-aware" simplification interface that highlights unsupported claims with evidence for end users.
- **Gap 3: Legal-domain NLI remains challenging, increasing the need for careful design and explainability.** Legal-domain NLI is difficult because premises may not explicitly state required information and may require domain knowledge; even contract-focused NLI work notes these challenges (Koreeda and Manning, 2021). Explainable NLI approaches in legal contexts further indicate the importance of providing explanations rather than only labels (Choi et al., 2023).

1.6 Contribution to the body of knowledge

1.6.1 Contribution to the research domain

This project proposes and implements an end-to-end, student-feasible pipeline that combines: (i) legal text simplification, (ii) claim extraction, and (iii) NLI-based verification, building on established NLI consistency ideas used in summarisation evaluation (Laban et al., 2022; Scirè et al., 2024) and the premise–hypothesis framing used in legal NLI datasets (Koreeda and Manning, 2021).

1.6.2 Contribution to the problem domain

For the Sri Lankan context, the project contributes a practical prototype that aims to improve safe comprehension of statute text by producing simplified output while surfacing "supported vs unsupported" highlights and statute evidence snippets. This aligns with Sri Lankan legal technology motivations around improving accessibility of legal information (Jayawardena et al., 2024; Fernando et al., 2024).

1.7 Research challenges

Challenge 1: Legal language complexity and long-range context.

Legal text contains complex vocabulary and structure, which complicates simplification and segmentation (Garimella et al., 2022; Cemri, Çukur and Koç, 2022).

Challenge 2: Hallucinations and factual inconsistency in generated rewrites.

LLMs can generate plausible but nonfactual content (Huang et al., 2025), and summarisation research documents hallucinations as a persistent risk (Maynez et al., 2020). Legal-domain evidence further suggests hallucination can be significant in legal tasks (Dahl et al., 2024).

Challenge 3: Legal-domain NLI difficulty and explainability.

Legal entailment may require implicit reasoning and domain knowledge, making verification hard (Koreeda and Manning, 2021). Providing interpretable verification signals is therefore important (Choi et al., 2023).

Challenge 4: Jurisdiction dependence and limited Sri Lanka–specific resources.

Legal NLP research highlights limited open datasets and jurisdiction dependence, which affects portability (Ariai, Mackenzie and Demartini, 2025; Akter et al., 2025).

1.8 Research questions

RQ1: How effectively can a baseline simplification module improve the readability of Sri Lankan statute sections while preserving meaning (Garimella et al., 2022; Cemri, Çukur and Koç, 2022)?

RQ2: To what extent can claim-level NLI verification detect unsupported statements in simplified statute outputs, using premise–hypothesis entailment relationships (Koreeda and Manning, 2021; Laban et al., 2022; Scirè et al., 2024)?

RQ3: Does presenting "status-aware" highlights and statute evidence snippets improve the practical trustworthiness of simplification outputs in a high-risk domain where hallucinations are a known concern (Huang et al., 2025; Dahl et al., 2024)?

RQ4: What evaluation protocol combining simplification quality and faithfulness/factuality measures is most suitable for a student-scale legal simplification system (Maynez et al., 2020; Laban et al., 2022; Scirè et al., 2024)?

1.9 Research aim

The aim of this research is to design, develop, and evaluate a hallucination-controlled legal text simplification framework for Sri Lankan statutes that verifies simplified claims against the original statute text using NLI-based entailment checking (Koreeda and Manning, 2021; Scirè et al., 2024).

1.10 Research objectives

RO1 (Specific, Measurable): Identify and document key challenges in legal text simplification and faithfulness for generated legal text, grounded in recent legal simplification and hallucination literature (Garimella et al., 2022; Huang et al., 2025).

RO2 (Achievable): Build a baseline simplification module that generates simplified versions of selected Sri Lankan statute sections, and record outputs for evaluation (Garimella et al., 2022).

RO3 (Measurable): Implement claim extraction to convert simplified outputs into atomic claims suitable for verification, following claim-oriented factuality evaluation ideas (Scirè et al., 2024).

RO4 (Measurable): Implement an NLI verification component that labels each claim as supported/unsupported using premise–hypothesis inference, aligned with legal NLI formulations (Koreeda and Manning, 2021).

RO5 (Relevant): Develop a web prototype that displays “status-aware” simplified text with evidence snippets, motivated by Sri Lankan needs for accessible legal context (Fernando et al., 2024; Jayawardena et al., 2024).

RO6 (Time-bound): Evaluate the system against baseline simplification using a defined evaluation protocol (automatic metrics plus human checking on a subset) and report results within the final semester, using established consistency evaluation approaches as guidance (Laban et al., 2022; Scirè et al., 2024).

1.11 Chapter summary

This chapter introduced the legal NLP context and the practical need for legal text simplification, especially for non-experts facing complex legal language (Garimella et al., 2022; Cemri, Çukur and Koç, 2022). It defined the central problem of simplifying Sri Lankan statute text while controlling hallucinations that are documented risks in text generation (Maynez et al., 2020; Huang et al., 2025). The chapter then established research gaps around Sri Lanka–specific statute simplification with integrated claim-level verification, stated the intended contributions and challenges, and presented the research questions, aim, and objectives.

CHAPTER 02: LITERATURE REVIEW

2.1 Chapter overview

This chapter critically reviews the body of knowledge underpinning faithful legal text simplification and hallucination control for Sri Lankan statutes. It begins with a conceptual mapping of the research area, then reviews the problem domain (legal text characteristics, simplification, hallucinations, and faithfulness). Next, it examines the enabling technologies stage-by-stage (pre-processing, modelling choices, verification methods, and evaluation). It then synthesises existing work through thematic analysis and evaluates current benchmarking practices. Finally, it identifies clear gaps that motivate the methodology presented in Chapter 3 (Ariai, Mackenzie and Demartini, 2025; Dehghani et al., 2025; Akter et al., 2025).

2.2 Concept map

This section presents a concept map that links the major ideas and sub-domains relevant to the project which are legal-language complexity, simplification, hallucinations/factuality, claim extraction, evidence alignment, NLI verification, and evaluation/benchmarking (Garimella et al., 2022; Huang et al., 2025; Scirè, Ghonim and Navigli, 2024).

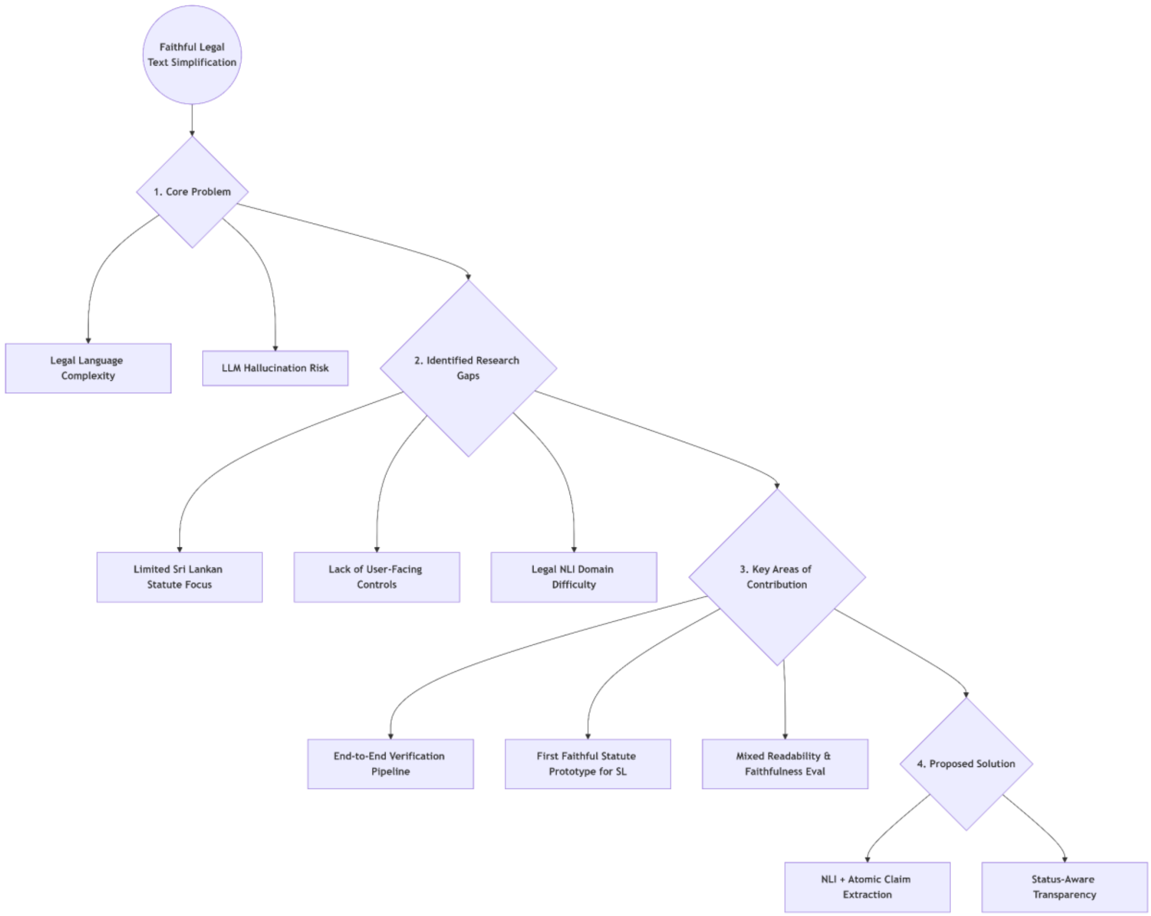

Figure 1: Concept Map of Research Domain.

2.3 Problem domain review

2.3.1 Introduction to the domain

Natural language processing (NLP) in the legal domain supports tasks such as legal search, question answering, summarisation, drafting, compliance, and decision support (Ariai, Mackenzie and Demartini, 2025; Dehghani et al., 2025). Legal texts introduce distinctive challenges compared with general-domain language, including long documents, dense structure, specialist vocabulary, and high stakes for correctness (Ariai, Mackenzie and Demartini, 2025; Dehghani et al., 2025).

Within this broad domain, legal text simplification aims to rewrite legal provisions into clearer language for non-experts while preserving legal meaning (Garimella et al., 2022; Cemri, Çukur and Koç, 2022). For statutes specifically, simplification must respect definitional dependencies, cross-references, exceptions, and formal constraints that directly affect interpretation (Muralidharan, 2021; Akter et al., 2025).

However, modern neural generation approaches can introduce hallucinations—content that appears plausible but is not supported by the source—creating a major reliability barrier in legal contexts (Maynez et al., 2020; Huang et al., 2025; Dahl et al., 2024). This motivates faithfulness control, where outputs are checked against source evidence to ensure claims remain supported (Maynez et al., 2020; Laban et al., 2022; Scirè et al., 2024).

A practical approach to faithfulness control is Natural Language Inference (NLI) verification, which determines whether a claim (hypothesis) is entailed by supporting text (premise) (Koreeda and Manning, 2021; Choi et al., 2023). This aligns with the project's core goal: producing simplified statute text that is status-aware (supported vs unsupported) via claim-level verification (Scirè, Ghonim and Navigli, 2024; Tang et al., 2025).

2.3.2 Key challenges

(1) Legal-language complexity and accessibility

Legal documents frequently contain long and nested sentences, specialised jargon, and complex discourse structure, making them difficult for laypeople to understand (Garimella et al., 2022; Cemri, Çukur and Koç, 2022). This complexity is particularly pronounced in contracts and statutory language where precision is prioritised over readability (Garimella et al., 2022; Muralidharan, 2021). Surveys of legal NLP emphasise that document length and structural complexity complicate segmentation, retrieval, modelling, and evaluation (Ariai, Mackenzie and Demartini, 2025; Akter et al., 2025). These accessibility barriers motivate simplification approaches, but simplifying statutes introduces a second challenge: ensuring outputs remain faithful to the source.

(2) Faithfulness and hallucination in generated legal text

Abstractive generation can produce statements not grounded in the source document, even when fluent, which is widely observed in summarisation and is directly relevant to simplification (Maynez et al., 2020). Hallucination in LLMs is commonly framed as generating "nonfactual" or "ungrounded" content, raising reliability concerns for real-world deployments (Huang et al., 2025). In legal settings, hallucinations are especially problematic because users may treat outputs as authoritative, amplifying harm (Dahl et al., 2024; Dehghani et al., 2025). Evidence of legal hallucinations in public-facing LLMs reinforces that domain-specific checking is required rather than assuming general correctness (Dahl et al., 2024; Magesh et al., 2025). To address hallucinations in simplification, the literature increasingly focuses on defining what faithfulness means and how it can be operationalised.

(3) Defining and operationalising "faithfulness" for simplification

Faithfulness is not just "similar wording"; it is the preservation of meaning and factual consistency with the source (Maynez et al., 2020). Legal simplification adds further constraints: simplified text must preserve obligations, conditions, exceptions and definitions that are legally operative (Garimella et al., 2022; Muralidharan, 2021). Recent work argues that benchmarks often focus on surface factual statements while overlooking reasoning-like "cognitive" statements, highlighting the need for stronger assessment standards (Tang et al., 2025). Finally, even well-defined faithfulness methods must be evaluated in the target legal system, making jurisdictional specificity essential.

(4) Jurisdictional specificity and gaps for Sri Lanka

Legal NLP systems and datasets are often jurisdiction-specific due to differences in institutions, terminology, and sources of law (Ariai, Mackenzie and Demartini, 2025; Curran et al., 2025). Sri Lanka has emerging legal NLP efforts such as embeddings and retrieval for Sri Lankan case-law question answering (Jayawardena et al., 2024) and constitution-focused chatbots (Fernando et al., 2024). However, these are not statute simplification systems, and they do not directly solve the "faithful simplification + verification" problem targeted here (Jayawardena et al., 2024; Fernando et al., 2024). Place-based evaluation work further supports that legal accuracy may vary across locations and legal systems, reinforcing the need to evaluate explicitly on Sri Lankan legal text rather than assuming transferability (Curran et al., 2025). Given these constraints, the following section reviews verification-style frameworks that can be adapted into a Sri Lanka-focused pipeline

2.3.3 Existing frameworks / standards / domain approaches

Within this project's scope, "frameworks" are best understood as evaluation and verification pipelines proposed in recent literature rather than external regulatory standards.

First, NLI-based consistency checking has been repeatedly studied as a way to detect factual inconsistencies in summarisation (Laban et al., 2022). However, earlier approaches struggled when applying sentence-level NLI models to document-level consistency problems, motivating aggregation approaches that better align premise granularity with document evidence (Laban et al., 2022).

Second, claim-based evaluation frameworks explicitly extract atomic claims and test them via entailment-style reasoning. For example, FENICE operationalises factuality evaluation using claim extraction and NLI to score whether generated summary claims are supported by the source (Scirè et al., 2024). This is directly aligned with the project's design choice to split simplified outputs into atomic claims and verify each claim against statute evidence (Scirè et al., 2024).

Third, recent benchmark proposals emphasise that "faithfulness hallucinations" are claims unsupported by provided context, and that evaluation should include both factual and reasoning-like statements (Tang et al., 2025). This motivates careful design of what counts as an "atomic claim" for statutes, including conditional and exception-bearing statements common in legislation (Tang et al., 2025; Garimella et al., 2022).

Finally, legal-domain NLI work highlights the mismatch between general-domain entailment models and legal entailment, motivating domain-specific modelling and explanation to support trust (Koreeda and Manning, 2021; Choi et al., 2023). Explainable NLI via text generation is particularly relevant because legal users benefit from transparent reasoning or evidence presentation when a claim is labelled unsupported (Choi et al., 2023).

2.3.4 Proposed architecture

This project positions itself at the intersection of (i) legal text simplification, (ii) hallucination/faithfulness research in generation, and (iii) NLI-based verification pipelines. The high-level conceptual architecture follows a generate → decompose → verify → present pattern: a baseline simplifier produces simplified statute text then the output is decomposed into atomic claims then each claim is verified using NLI where the premise is an evidence chunk from the original statute and the user receives status-aware simplified text with highlights and evidence snippets (Maynez et al., 2020; Laban et al., 2022; Scirè et al., 2024).

Sri Lankan law is incorporated directly through the statute corpus used as both the input source and the ground-truth evidence base. Rather than relying on external retrieval to generate content, the system simplifies a statute provision selected from (or pasted from) the Sri Lankan statute corpus and then verifies each simplified claim against the same provision using claim extraction and NLI. This ensures that outputs are evaluated and presented within the target jurisdiction, addressing the concern that legal NLP performance and reliability do not automatically transfer across legal systems and that local legal text should be the basis of evaluation (Ariai, Mackenzie and Demartini, 2025; Akter et al., 2025; Curran et al., 2025; Jayawardena et al., 2024).

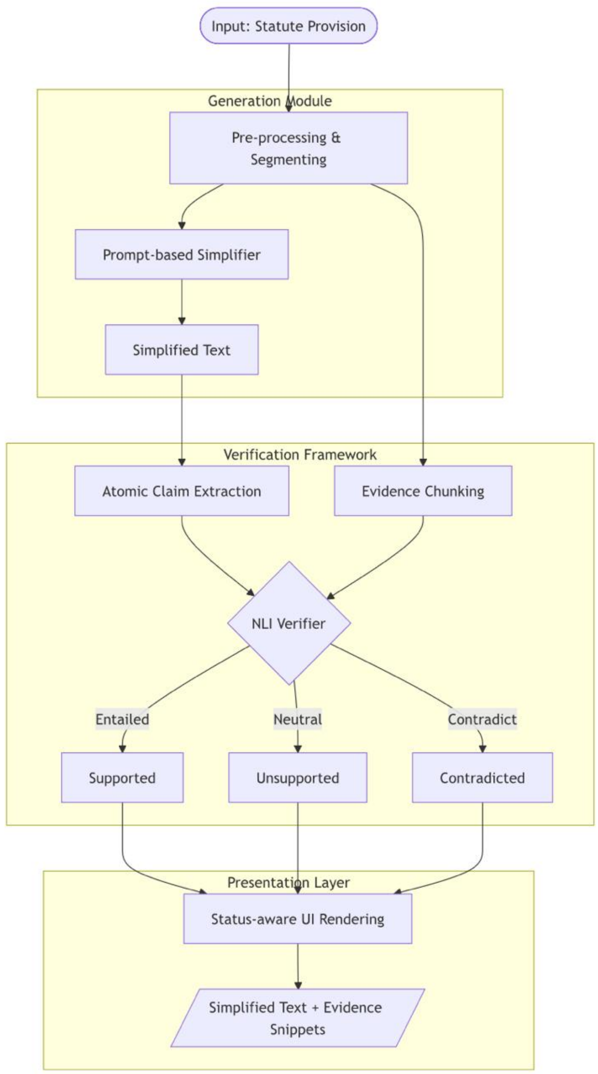

Figure 2: Proposed System Architecture

2.4 Technological review

2.4.1 Data acquisition & pre-processing

Legal-domain surveys highlight that legal documents are typically long and structurally complex, requiring careful segmentation and representation before modelling (Ariai, Mackenzie and Demartini, 2025; Akter et al., 2025). For statutes, pre-processing commonly includes section-level splitting, sentence segmentation, and chunking strategies to enable downstream retrieval or verification (Scirè et al., 2024).

When verification depends on matching claims to evidence, chunking is not simply a performance optimisation; it shapes whether the premise contains sufficient context to entail a claim (Laban et al., 2022). Work revisiting NLI inconsistency detection specifically identifies premise granularity mismatch as a key failure mode and proposes segmentation + aggregation to improve reliability (Laban et al., 2022). Sri Lankan legal NLP work has demonstrated retrieval/embedding approaches for local legal texts in QA settings (Jayawardena et al., 2024), but this project uses evidence selection only within the user-provided statute section. For verification, the statute section is segmented into chunks and the most relevant chunk(s) are selected as NLI premises, because premise granularity affects inconsistency detection performance (Laban et al., 2022).

2.4.2 Model architectures

- **Simplification module.** Legal text simplification is studied using a range of approaches, including domain adaptation, lexical/syntactic simplification, and summarisation-style rewriting (Garimella et al., 2022; Cemri, Çukur and Koç, 2022). A statutes-focused thesis demonstrates the feasibility of applying automatic text simplification to statutory language, reinforcing the relevance of statute-specific constraints and evaluation needs (Muralidharan, 2021). Hybrid approaches in adjacent jurisdictions also combine legal-domain encoders with rewriting/summarisation methods to improve readability while attempting to preserve meaning (Sharma et al., 2025).
- **Verification module (NLI).** NLI formalises verification as entailment classification between a premise and hypothesis, making it suitable for claim-level checking (Koreeda and Manning, 2021). ContractNLI proposes document-level NLI for contracts, reflecting that legal inference often depends on context spread across long documents rather than

single sentences (Koreeda and Manning, 2021). Legal-domain explainable NLI further argues that typical pre-trained approaches can underperform in legal contexts, motivating legal-specific adaptation and explanation generation to improve usability and trust (Choi et al., 2023).

- **Faithfulness/consistency checking for generation.** NLI-based inconsistency detection for summarisation has been revisited with more effective aggregation methods, demonstrating that NLI can become competitive when the document–sentence granularity mismatch is addressed (Laban et al., 2022). Claim-extraction + NLI pipelines (e.g., FENICE) provide an explicit blueprint for evaluating whether generated content is supported by the source, aligning closely with a verification-first simplification design (Scirè et al., 2024).

2.4.3 Hyperparameter tuning / calibration

This project's feasibility constraints suggest minimising heavy training while still enabling calibration and thresholding of verifier outputs. Hallucination surveys note that mitigation often combines multiple layers (prompting, retrieval grounding, verification, and post-processing) rather than relying on a single mechanism (Huang et al., 2025; Tonmoy et al., 2024).

Where fine-tuning is feasible, literature supports domain-specific adaptation for legal tasks (Ariai, Mackenzie and Demartini, 2025; Koreeda and Manning, 2021). However, careful design can remain effective even without large-scale training by treating verification as a pipeline component with tunable thresholds and explainable outputs (Laban et al., 2022; Choi et al., 2023).

Because legal hallucinations can persist even in widely used systems, calibration should be evaluated empirically rather than assumed (Dahl et al., 2024; Magesh et al., 2025).

2.4.4 Evaluation metrics used currently

A recurring theme across summarisation and simplification research is that surface similarity metrics do not directly measure factual consistency or faithfulness (Maynez et al., 2020; Akter et al., 2025). Faithfulness-focused work shows that hallucinations can be frequent even when text appears coherent, motivating explicit factuality/consistency assessment (Maynez et al., 2020; Huang et al., 2025). Standard overlap-based metrics such as ROUGE-L, which measures the longest common subsequence between a generated output and a reference, are widely used as baseline measures of content coverage in summarisation and simplification research (Maynez et al., 2020; Akter et al., 2025). However, ROUGE-L does not capture semantic equivalence or factual correctness, meaning a simplified output may score well on ROUGE-L while still introducing unsupported content. This limitation motivates the complementary use of BERTScore, which computes token-level similarity using contextual embeddings from pre-trained language models, providing a semantics-aware measure of similarity between generated and reference text. While BERTScore improves over surface overlap, it still does not directly penalise hallucinated claims, which is why claim-level NLI verification is used as a third evaluation layer in this project.

NLI-based factuality metrics operationalise evaluation by testing entailment between extracted claims and source evidence (Laban et al., 2022; Scirè et al., 2024). FENICE in particular combines claim extraction and NLI to evaluate factuality, while also discussing limitations of emerging metrics, implying that multiple evaluation views may be necessary (Scirè et al., 2024).

For legal-domain hallucination profiling and tool reliability, empirical studies emphasise measuring factual inconsistency and erroneous legal output rather than relying on subjective impressions of fluency (Dahl et al., 2024; Magesh et al., 2025). Place-based legal evaluation further introduces comparative "hallucination rates" across jurisdictions, supporting the need for jurisdiction-aware benchmarking rather than generalising across places (Curran et al., 2025).

2.5 Existing work

2.5.1 Thematic grouping

Theme A: Legal text simplification methods and datasets

Legal-domain simplification research highlights data scarcity, the challenge of preserving meaning, and the need for specialised approaches due to legal structure and vocabulary (Garimella et al., 2022; Cemri, Çukur and Koç, 2022). Statute-focused simplification work demonstrates practical pipelines tailored to statutory drafting style, supporting this project's focus on Sri Lankan statutes rather than general legal text (Muralidharan, 2021). Adjacent jurisdiction work (e.g., Indian legal simplification) indicates that hybrid architectures and legal-domain encoders are actively explored, but they do not inherently guarantee faithfulness without verification (Sharma et al., 2025).

Theme B: Hallucination and reliability in legal AI systems

General LLM hallucination surveys describe hallucination as a systematic reliability issue and review detection and mitigation strategies (Huang et al., 2025; Tonmoy et al., 2024). Legal-focused studies provide empirical evidence that public-facing models and tools can hallucinate legal “facts”, which is especially risky given user trust and the consequences of legal misinformation (Dahl et al., 2024; Magesh et al., 2025). Place-based legal evaluation further suggests that accuracy can vary based on jurisdiction, motivating evaluations grounded in the target legal system (Curran et al., 2025).

Theme C: Faithfulness and factuality evaluation for generated text

Foundational work on abstractive summarisation shows that hallucinations can be substantial and that standard training/decoding can encourage unsupported content, implying that simplification (also a generation task) can inherit these issues (Maynez et al., 2020). NLI-based inconsistency detection and claim-based evaluation frameworks represent a concrete technical direction for identifying unsupported statements (Laban et al., 2022; Scirè et al., 2024). Benchmark design work also argues that evaluation should capture reasoning-like claims, relevant to statutes where meaning often depends on conditions and exceptions (Tang et al., 2025).

Theme D: Legal-domain NLI and explainability

ContractNLI formalises legal NLI at document level and frames it as a real-world application that reflects legal-document complexity and practical cost barriers (Koreeda and Manning, 2021). Explainable legal NLI via text generation supports the need for transparent justifications, which aligns with this project’s UI design of showing evidence snippets and highlighting unsupported claims (Choi et al., 2023).

Theme E: Sri Lankan legal NLP systems (context and limitations)

Sri Lanka-specific research includes retrieval/embedding systems for case law QA (Jayawardena et al., 2024) and constitution chatbots aimed at improving legal awareness (Fernando et al., 2024). These systems demonstrate increasing feasibility of Sri Lanka-specific legal NLP, but they do not target statute simplification nor do they integrate claim-level faithfulness verification, leaving a clear gap (Jayawardena et al., 2024; Fernando et al., 2024).

2.5.2 Cross-theme linking

Across themes, a consistent pattern emerges: (1) legal simplification is motivated by accessibility and complexity, but meaning preservation remains hard (Garimella et al., 2022; Cemri, Çukur and Koç, 2022); (2) hallucination is a recognised failure mode in LLM generation and is empirically observed in legal systems and tools (Huang et al., 2025; Dahl et al., 2024; Magesh et al., 2025); and (3) NLI-based and claim-based verification provides an actionable pathway to operationalise faithfulness at the statement level (Laban et al., 2022; Scirè et al., 2024).

Sri Lankan legal NLP is progressing, but primarily in QA and retrieval rather than faithful statute simplification, strengthening the rationale for a Sri Lanka-focused verification pipeline and prototype (Jayawardena et al., 2024; Fernando et al., 2024).

2.6 Evaluation and benchmarking

Legal NLP surveys emphasise that datasets and evaluation practices strongly shape what models learn and how claims of effectiveness should be interpreted (Ariai, Mackenzie and Demartini, 2025; Dehghani et al., 2025). In legal summarisation, surveys highlight diversity in datasets, summarisation targets, and evaluation protocols, which makes direct comparisons difficult and motivates careful benchmark selection (Akter et al., 2025).

Limitations of common evaluation practices. Faithfulness research demonstrates that generation outputs can score well on conventional metrics while still containing unsupported content (Maynez et al., 2020). This undermines evaluation strategies that rely only on similarity-based metrics and supports integrating factuality checks (Maynez et al., 2020; Scirè, Ghonim and Navigli, 2024).

NLI/claim-based benchmarking direction. SUMMAC provides evidence that NLI can be effective for inconsistency detection when the evaluation pipeline addresses document-level evidence aggregation (Laban et al., 2022). FENICE proposes factuality evaluation using claim extraction and NLI, closely matching the project's idea of atomic-claim verification for simplified statute text (Scirè et al., 2024). CogniBench further argues that evaluation should cover both factual and reasoning-like statements, which is relevant to statutes where a “faithful simplification” must preserve conditions, exceptions, and definitional dependencies (Tang et al., 2025).

Jurisdiction-aware benchmarking. Place-based comparisons of legal hallucination rates reinforce that evaluation must be grounded in the intended legal environment and query distribution, rather than assuming uniform model capability across legal systems (Curran et al., 2025). Sri Lankan legal QA research also demonstrates practical retrieval evaluation considerations using local case-law corpora and ranking pipelines (Jayawardena et al., 2024).

Together, these findings motivate the project’s evaluation approach which is to assess readability/simplicity improvements while directly measuring claim support using NLI verification against the source statute text, and report results specifically on Sri Lankan statutory content (Maynez et al., 2020; Laban et al., 2022; Scirè et al., 2024).

2.7 Chapter summary

This chapter reviewed the literature on legal text simplification, hallucinations and faithfulness in generation, NLI-based verification, and Sri Lanka-specific legal NLP. The evidence shows that legal language presents substantial complexity for non-experts, that hallucination is a recognised and empirically observed failure mode in LLM-generated legal outputs, and that claim-based + NLI verification pipelines provide a concrete mechanism for detecting unsupported statements. Current Sri Lankan legal NLP work demonstrates momentum in retrieval and QA, but does not address faithful statute simplification with claim-level verification. These gaps directly motivate the methodology in Chapter 3, which designs a feasible generate–verify pipeline and an evaluation plan grounded in Sri Lankan statutes (Garimella et al., 2022; Huang et al., 2025; Laban et al., 2022; Jayawardena et al., 2024; Scirè, Ghonim and Navigli, 2024).

CHAPTER 03: METHODOLOGY

3.1 Chapter overview

This chapter explains how the project will be designed, developed, and evaluated in a way that is feasible for a single student while producing an academically defensible prototype. The methodology is structured around (1) a prototype-driven research strategy, (2) an Agile development approach, and (3) a mixed evaluation plan measuring readability and faithfulness (Maynez et al., 2020; Laban et al., 2022). This chapter details the technical implementation steps, including claim-level Natural Language Inference (NLI) verification to detect unsupported statements (Scirè et al., 2024).

3.2 Research methodology (Saunders' Research Onion)

Table 1: Saunders' Research Onion for this project

Layer	**Choice in this project**	**Justification**
Philosophy	Pragmatism	The project combines measurable system metrics with human judgement of meaning preservation, which suits a practical system-building study.
Approach	Deductive	The system tests whether claim-level verification improves faithfulness of simplified outputs compared with simplification alone.
Strategy	Prototype + experiment	A working web prototype is developed and evaluated under controlled conditions using a fixed statute subset.
Methodological choice	Mixed-methods	Automatic readability + supportedness metrics are complemented with human checking for meaning preservation.
Time horizon	Cross-sectional	The evaluation is conducted on a defined dataset subset at a single point in time near project completion.

Techniques/procedures	Prompt-based simplification → deterministic evidence span generation (P1–Pn) → atomic claim extraction → evidence span selection → NLI verification → UI highlighting	Matches factuality evaluation pipelines using claim decomposition and NLI checking.

This Research Onion configuration (Saunders, Lewis and Thornhill, 2019) is suitable for this project because it supports iterative prototyping while ensuring the final output is assessed using measurable outcomes. The emphasis on claim-based verification is justified by prior work showing hallucinations and factual inconsistency in generation tasks (Maynez et al., 2020) and by research demonstrating that NLI-based inconsistency detection and claim-based factuality evaluation can provide structured signals of supportedness (Laban et al., 2022; Scirè et al., 2024).

3.3 Development methodology

An Agile approach will be used because the project integrates multiple NLP components (prompt-based simplification, claim extraction, and NLI verification) that require iterative testing. Iterative cycles support early prototyping and incremental improvements to prompts, claim extraction heuristics, and verification thresholds (Scirè et al., 2024). Work is organized into 2 week sprints. This is essential because factuality control methods often require repeated calibration before the pipeline becomes stable enough for final evaluation (Maynez et al., 2020).

3.3.1 System design and implementation pipeline

The ClearClause system implements a seven-step pipeline following a generate → decompose → verify → present architecture. Verification is grounded strictly within the same statute provision provided as input; no external retrieval is used.

Step 1: Input acquisition. The user enters or selects a Sri Lankan statute provision from the Penal Code or the Maintenance Act No. 37 of 1999. PDF upload is a secondary, optional enhancement.

Step 2: Deterministic evidence span generation. The statute provision is converted into labelled evidence spans P1–Pn using deterministic rule-based segmentation. Segmentation is driven by legal drafting markers including subsection boundaries, provisos, exceptions and sentence boundaries. These spans preserve the structure of the legal provision and serve as premise candidates for NLI verification (Laban et al., 2022; Koreeda and Manning, 2021).

Step 3: Prompt-based simplification. A prompt-based LLM simplification baseline generates a plain-language version of the provision using instruction-style prompts that explicitly instruct the model to preserve obligations, permissions, conditions, exceptions, definitions, numbers and penalties, and to add no information not present in the source (Garimella et al., 2022; Sharma et al., 2025).

Step 4: Atomic claim extraction. The simplified output is decomposed into short, independently checkable legal claims. Each atomic claim captures a single obligation, permission, condition, exception or definition to allow precise, claim-level verification (Scirè et al., 2024).

Step 5: Evidence span selection. Each extracted claim is matched against the most relevant P1–Pn evidence spans from the original provision. Premise granularity is a known factor in NLI inconsistency detection performance, and selecting the most relevant span rather than the full document reduces noise in the verification step (Laban et al., 2022).

Step 6: NLI verification. A DeBERTa-based NLI verifier classifies the relationship between each selected evidence span (premise) and each atomic claim (hypothesis), producing one of three labels: entailment (supported), neutral (unsupported), or contradiction (contradicted) (Koreeda and Manning, 2021; Choi et al., 2023).

Step 7: Status-aware UI rendering. The web interface renders the simplified output with colour-coded claim highlights distinguishing supported, unsupported, and contradicted claims. Each claim is paired with its matched evidence span, displayed as a clickable evidence snippet to improve transparency for the user.

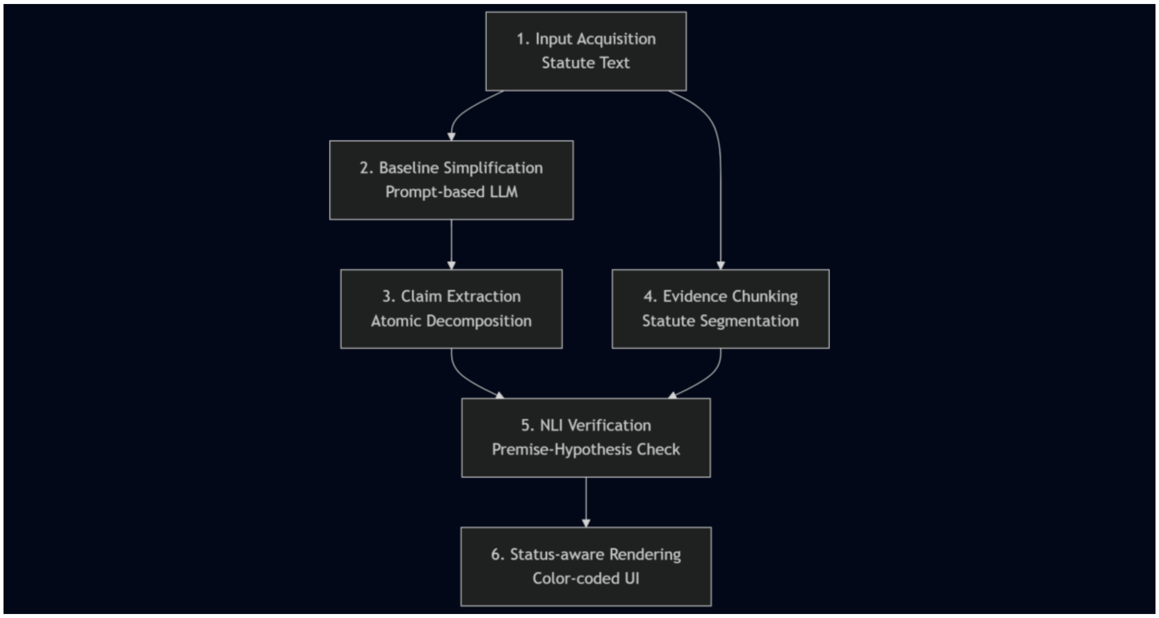

Figure 3: Implementation Pipeline

3.3.2 Evaluation design

Evaluation is designed to measure readability and faithfulness simultaneously:

- **Baselines:** Baseline A (Simplification only) is compared against the Proposed System (Simplification + NLI verification).
- **Readability Metrics:** Automated scores such as FKGL and ARI are used to quantify simplicity improvements (Muralidharan, 2021). ROUGE-L and BERTScore are additionally used to measure content overlap and semantic similarity between simplified and source text, complementing the readability measures.
- **Faithfulness Metrics:** Claim-supportedness rate (percentage of claims labelled entailment vs neutral/contradiction) is reported using the NLI verifier, following NLI-based inconsistency detection and claim-based factuality evaluation approaches (Laban et al., 2022; Scirè et al., 2024).
- **Human Review:** A small subset (e.g., 10–20 provisions) will undergo manual review using a simple rubric (meaning preserved / minor drift / major drift), focusing on obligations, conditions, exceptions and definitions (Garimella et al., 2022; Cemri, Çukur and Koç, 2022).

3.4 Project management methodology

3.4.1 Schedule

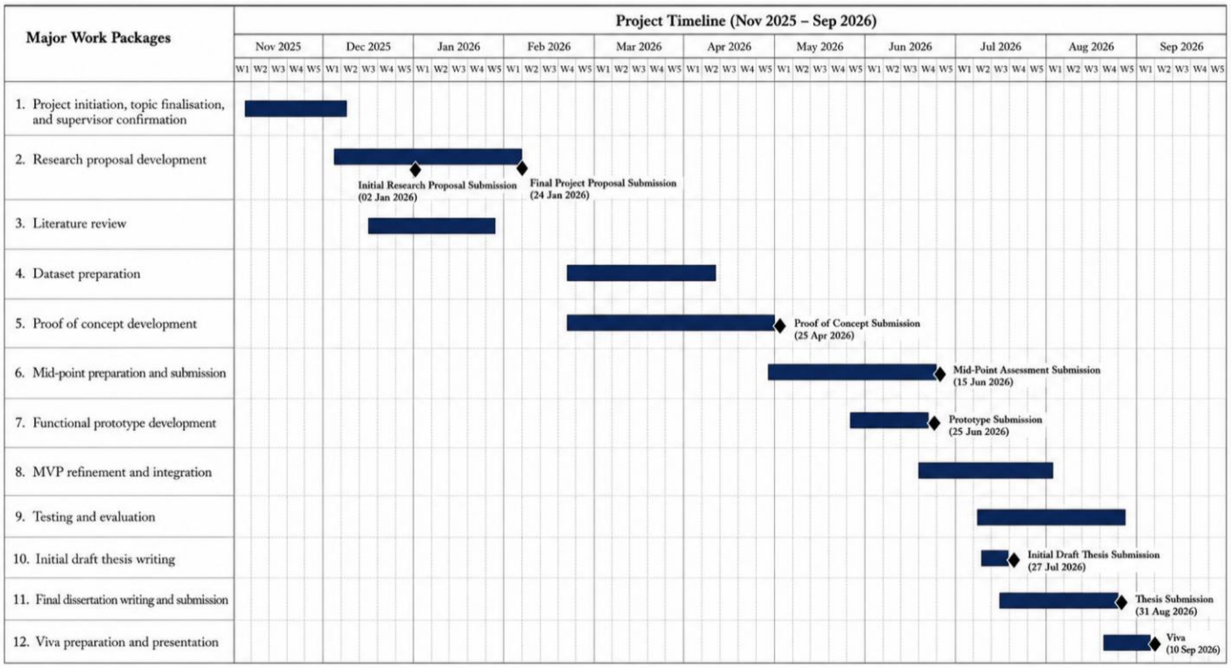

Figure 4: Gantt Chart

3.4.2 Deliverables and dates

Table 2: Deliverables and Deadlines

Deliverable	Description	Due Date (End of Month)
Project Proposal	Finalised and authorised study proposal including project goals, scope, preliminary literature review on legal text simplification, hallucinations/factuality, NLI verification, and the proposed methodology and evaluation plan.	Jan-26
Dataset Preparation	Sri Lankan statute corpus prepared in clean text format with metadata (Act name, year,	Mar-26

	section/subsection). Evaluation subset selected and documented for reproducible testing.	
Proof of Concept	Working proof of concept demonstrating the end-to-end core flow on a small sample: statute input → prompt-based simplification → claim extraction → NLI verification → supported/unsupported labels.	Apr-26
Functional Prototype	Functional prototype with a basic web interface and integrated pipeline: simplification baseline + claim-level NLI verification + evidence chunking/selection from the same statute provision, producing status-aware output.	Jun-26
Minimum Viable Product (MVP)	Improved MVP with stable UI highlighting, evidence display per claim, verification summary panel, and automated computation of readability metrics (FKGL, ARI), semantic similarity metrics (ROUGE-L, BERTScore), and claim-supportedness rates.	Jul-26
Testing & Evaluation Report	Complete evaluation results on the defined subset: readability comparison (baseline vs proposed), supportedness distribution, and manual meaning-preservation review outcomes, including analysis and refinement notes.	Aug-26
Final Dissertation	Full dissertation submitted, including implementation details, evaluation findings, discussion, limitations, and future work, aligned with the FYP structure requirements.	Sep-26
Viva and Presentation	Final oral presentation and defence of the project, including live demo of the prototype and summary of evaluation outcomes.	

3.4.3 Resources

- **Hardware resources** Development requires a standard laptop for prototype development and running smaller inference workloads. Google Colab is used for training and evaluation.
- **Software resources** The implementation stack includes Python, Hugging Face Transformers, PyTorch, and a FastAPI with a web UI.
- **New technical skills** The project requires practical implementation skills in claim decomposition and NLI-based verification pipelines, including evidence chunking and entailment-based supported scoring.

3.4.4 Data requirements

The project uses a structured collection of Sri Lankan statutory provisions from the Penal Code and the Maintenance Act No. 37 of 1999, stored as clean text with metadata (Act name, year, section/subsection identifiers). Legal NLP surveys emphasise that corpus structuring and dataset handling shape system design and evaluation (Ariai, Mackenzie and Demartini, 2025; Akter et al., 2025). Although this project does not implement retrieval-augmented generation (RAG), a Sri Lankan statute corpus is still essential as the project's primary legal data source and evaluation domain. The corpus provides structured statute provisions that can be selected within the prototype to populate the input provision, enabling consistent experimentation and traceability of outputs back to the exact legal source. Verification is grounded strictly in the selected provision: the original provision is converted into labelled evidence spans P1–Pn using deterministic rule-based segmentation, and the most relevant span(s) are selected as NLI premises for each extracted claim prior to classification.

3.4.5 Risks and mitigation

Table 3: Risk and Mitigation Table

Risk Event	Likelihood	Impact	Mitigation Strategy
Meaning Drift	High	Critical	Use strict simplification prompts that explicitly instruct the model to preserve SHALL/MAY/SHALL NOT constructs, obligations, conditions, exceptions, definitions

			and penalties; verify each atomic claim against its matched evidence span using NLI (Scirè et al., 2024).
NLI Mismatch	Medium	High	Use a DeBERTa-based NLI verifier; calibrate confidence thresholds empirically on manually labelled legal claim pairs; keep uncertain (neutral) claims visible in the UI rather than suppressing them (Koreeda and Manning, 2021; Choi et al., 2023).
Data Scarcity	High	Medium	Build a small controlled evaluation dataset from selected Penal Code and Maintenance Act No. 37 of 1999 provisions, with manual simplifications and NLI labels, to enable reproducible testing independent of large-scale annotated corpora.

3.5 Chapter summary

An outline of the intended research methodology for this project has been provided in this chapter. The study is positioned as a pragmatic and deductive research project supported by prototype-based development and experimental evaluation. Agile development is used to support iterative progress across dataset preparation, model testing, frontend development and backend pipeline implementation. The proposed ClearClause pipeline follows a generate, decompose, verify and present structure, where Sri Lankan statute text is simplified, broken into atomic claims, linked with source evidence and prepared for NLI-based verification. The selected evaluation approach combines readability metrics, semantic similarity metrics, claim-supported checking and human review, creating a practical foundation for the later implementation and assessment stages of the project.

CHAPTER 04: LEGAL, ETHICAL, SOCIAL AND PROFESSIONAL ISSUES

4.1 Chapter overview

This chapter discusses the legal, ethical, social and professional issues connected to the ClearClause legal text simplification system. Since the project focuses on simplifying Sri Lankan statutory provisions using large language models, it was important to consider several risks. These included incorrect legal interpretation, hallucinated outputs, data protection concerns, user over-reliance and professional responsibility.

ClearClause was developed as an academic prototype. It was not intended to replace professional legal advice.

4.2 Legal issues

One of the main legal issues in this project was the risk that users might mistake the simplified output for legal advice. ClearClause was designed to convert statutory text into plain English. However, it was not designed to provide legal advice, legal interpretation or recommendations based on a user's personal situation.

This distinction was important because legal advice requires professional judgement and a proper understanding of the user's specific facts. For this reason, the frontend included a legal disclaimer banner. The purpose of the disclaimer was to clearly inform users that the output was intended only for educational and informational purposes.

Another important legal issue was the possibility of changing the meaning of the law during simplification. Legal provisions often use very precise wording, and even small changes can affect their legal meaning. For example, changing "shall" to "may", removing an exception or simplifying a punishment incorrectly could mislead the user. To reduce this risk, strict simplification rules were used during dataset construction. The simplified text had to preserve obligations, conditions, exceptions, numerical values, punishments and legal roles.

Data protection was also considered as a background concern. The core dataset used public statutory text, and the main system did not require or collect personal user data for simplification. Sri Lanka's Personal Data Protection Act No. 9 of 2022 provides a legal framework for regulating the processing of personal data and protecting data subjects in Sri Lanka (Parliament of Sri Lanka, 2022). The Data Protection Authority of Sri Lanka also provides official information and guidance on the Act and personal data processing in Sri Lanka (Data Protection Authority of Sri Lanka, 2024). In addition, the ICO's guidance on AI and data protection highlights the importance of fairness, transparency and compliance with data protection requirements in AI systems (ICO, 2023).

Intellectual property and source acknowledgement were also important considerations. The project used statute texts, academic research papers, open-source libraries and pretrained models. All sources were properly cited in the report. The statutory provisions used in the dataset were also stored with identifiers such as the statute name and section number, allowing each provision to be traced back to its original legal source.

4.3 Ethical issues

The main ethical issue in this project was hallucination. In this context, hallucination refers to the risk that the model may generate information that is not supported by the original statute. This was a serious concern because users may trust the simplified output if it is written clearly and confidently. Previous research on text generation has shown that generated outputs can include unsupported or unfaithful content, even when the writing appears fluent and readable (Maynez et al., 2020; Huang et al., 2025). In a legal context, this risk is even more serious because incorrect information could affect how a person understands rights, duties, offences or penalties.

To reduce this issue, the project was designed around hallucination control. The system did not only generate a simplified version of the statute. It also included a verification framework using source spans, atomic claim extraction, evidence linking and NLI verification. This design aimed to check whether the simplified claims were supported by the original statute text.

Meaning preservation was another important ethical concern. A simplification system should not make the law easier to read by removing important legal details. During dataset construction, the simplified text was written to preserve legal duties, exceptions, conditions, punishments, numbers, dates and legal roles. This was important because an output that is simple but legally inaccurate would not be useful or responsible.

Transparency was also considered. The system was designed to show source spans and evidence links, rather than only presenting a simplified paragraph. This was important because users and evaluators needed to see how each simplified claim connected back to the original statute. UNESCO's Recommendation on the Ethics of Artificial Intelligence emphasises transparency, explainability, traceability and human oversight as important principles for ethical AI systems (UNESCO, 2021).

Human review was included because automatic metrics and AI models can still make mistakes. For example, FKGL and ARI can show whether text is easier to read, but they cannot prove that the legal meaning has been preserved. Similarly, BERTScore and ROUGE-L can compare similarity, but they cannot fully identify legal errors. Therefore, the project used human simplified reference text and NLI verification to support a more reliable evaluation process.

4.4 Social issues

The project has a positive social purpose because it aims to make legal information easier to understand. Sri Lankan statutes can be difficult for non-lawyers because they often use formal legal language, long sentences and complex conditions. By simplifying statutory provisions into plain English, the project supported legal awareness and improved basic access to legal information.

At the same time, there was also a risk of over-reliance on AI-generated explanations. Users may assume that the simplified output is completely correct simply because it appears clear and confident. This risk is especially important in legal contexts, where users may make decisions based on what they understand from the output. To reduce this issue, the interface included a legal disclaimer, and the system design focused on evidence display and verification rather than plain text generation alone.

Accessibility was another social issue. The prototype was built in English and used a web-based interface. This means it may not fully support users who are more comfortable in Sinhala or Tamil, users with lower digital literacy or users without reliable internet access. For this reason, the midpoint version of the project was presented as an academic prototype, rather than as a complete public legal access platform.

Dataset coverage was also a limitation. The project used the Penal Code and the Maintenance Act, but Sri Lankan law includes many other statutes and legal areas. Therefore, the results of this project cannot be generalised to all Sri Lankan legal text without further testing. This limitation was important to acknowledge because an AI system should not be presented as more reliable or more complete than it actually is.

4.5 Professional issues

Professional responsibility was important because this project involved the use of AI in a sensitive legal domain. The BCS Code of Conduct highlights professional duties such as working in the public interest, maintaining competence, acting with integrity and behaving responsibly as a computing professional (BCS, 2025). These duties were relevant because an AI-based legal simplification system could influence how users understand legal information.

The public interest principle was addressed by designing the system to improve access to legal information while also reducing potential harm. The project did not present the system as a legal advisor. Instead, it was positioned as a legal text simplification and verification prototype. The use of disclaimers, source spans and planned NLI verification helps reduce the risk of misleading users.

Professional competence was also important. The project clearly explained what had been completed and what had not been completed at the midpoint stage. For example, the project had completed legal data collection, dataset construction, baseline model evaluation, frontend prototyping, span construction and keyword-based evidence linking. However, the DeBERTa-based NLI verification stage had not been fully completed at that point. Reporting this honestly was necessary to avoid overstating the system's capabilities.

Integrity was followed by clearly documenting the dataset preparation process, model evaluation process and implementation limitations. The baseline evaluation compared multiple models instead of assuming that one model was automatically the best. Qwen 3 8B was selected based on the current baseline comparison, not simply because of personal preference. This made the model selection process more transparent.

The project also followed academic integrity requirements. All academic papers, legal sources, models, libraries and tools used in the project were properly referenced. The screenshots, outputs and datasets included in the report represented actual project work.

4.6 Chapter summary

The legal, ethical, social and professional issues connected to the ClearClause project have been discussed in this chapter. The main legal concern was the risk that users may treat simplified statutory text as legal advice, which was addressed through the use of disclaimers and clear system positioning. Ethical concerns such as hallucination, meaning drift, transparency and human review were also considered because the system works in a sensitive legal domain. The chapter also discussed social issues such as access to legal information, user over-reliance, English-only limitations and limited statute coverage. Professional responsibilities were addressed through public interest, competence, integrity and honest reporting of the project's current limitations.

CHAPTER 05: IMPLEMENTATION AND DATA COLLECTION

5.1 Chapter overview

This chapter presents the implementation and data collection work completed for the ClearClause system up to the midpoint stage. It explains how the Sri Lankan legal data was collected, how the dataset was prepared, how the baseline models were evaluated, and how the frontend and backend were structured.

5.2 Technology stack used

The implementation uses a simple Python-based NLP stack together with a lightweight web prototype. Python was chosen because most of the text processing, model evaluation and metric calculation work can be managed more easily using Python libraries. Google Colab was used for the baseline model evaluation experiments.

The frontend was built using HTML, Tailwind CSS and JavaScript. The main purpose of the interface was to accept a legal provision, send it to the backend and display the simplified output together with claims and source spans.

FastAPI was used for the backend because it works well with Python and makes it easier to expose each stage of the pipeline as a separate endpoint. This also helped with testing, since both the full pipeline and the individual debugging endpoints could be tested separately through the FastAPI documentation page.

Table 4: Technology stack used

Component	Tool or technology	Purpose
Programming language	Python	Data preparation, model evaluation and backend pipeline
Notebook environment	Google Colab	Running baseline model experiments
Frontend structure	HTML	Creating the ClearClause web pages
Frontend styling	Tailwind CSS	Styling the user interface quickly

Frontend interaction	JavaScript	Sending requests and displaying backend responses
Backend framework	FastAPI	Creating API endpoints for the pipeline
Model library	Hugging Face Transformers	Loading and running transformer models
Deep learning framework	PyTorch	Supporting model inference
Data handling	pandas	Reading and writing CSV datasets
Readability metrics	FKGL and ARI	Measuring readability of generated outputs
Semantic similarity metrics	ROUGE-L and BERTScore F1	Comparing generated outputs with reference simplifications

5.3 Legal data collection

The project uses Sri Lankan statutory text as its main data source. Two statutes were selected for the dataset: the Penal Code and the Maintenance Act.

The Penal Code was used as the main dataset because it contains different types of legal provisions, including offences, punishments, exceptions, conditions and criminal liability rules. This made it useful for testing whether a simplification system could handle complex legal language without changing the legal meaning.

The Maintenance Act was added as a second statute so that the project did not rely only on criminal law provisions. It contains a different type of legal language, mainly related to maintenance duties, family relationships and legal responsibilities.

Both statutes were obtained as PDF documents. Since PDF extraction does not always produce clean text, the extracted data went through a separate review and cleaning process before being used in the dataset.

5.4 PDF extraction and preprocessing

The statute PDFs were processed using Python scripts. The extraction process produced separate outputs so the raw text could be reviewed before being added to the final dataset. For the Penal Code, the extraction process produced a page-level dataset, a cleaned text file and a candidate provision dataset.

The page-level dataset stored the extracted text page by page. This was useful because it made it easier to check where a particular piece of text came from if an extraction issue appeared. The cleaned text file contained the full statute text after obvious formatting problems were removed. The candidate provision dataset contained possible legal provisions that could be used for simplification.

Several extraction problems were identified during dataset construction because of formatting inconsistencies in the PDF documents. Some provisions were cut off, while others were incomplete because of page boundaries. In some cases, headings were merged into the provision text. There were also very short fragments and PDF formatting artefacts that were not useful for modelling. These issues showed that PDF extraction could not be trusted without manual review.

Because of this, a quality review step was added before the final dataset was created. Problematic rows were removed so that incomplete or misleading legal text did not enter the simplification dataset. This was important because the model output and evaluation depended heavily on the quality of the source text.

example_i	act_name	section_id	unit_label	unit_type	page_start	source_tex	has_modal	has_excep	has_numb	has_roles	needs_con	bad_fragm	simple_tex	split	notes
SL00001	Penal Code	Section 6		section_body		6. Every ex	0	0	1	0	1	0			
SL00002	Penal Code	Section 7		section_bo	4	7. The pror	0	1	1	1	0	0			
SL00003	Penal Code	Section 8		section_body		8. importin	0	0	1	0	0	0			
SL00004	Penal Code	Section 9		section_body		9. The word	0	0	1	1	0	0			
SL00005	Penal Code	Section 10		section_body		10. The wo	0	0	1	0	0	0			
SL00006	Penal Code	Section 11		section_body		11. Republ	0	0	1	0	0	1			
SL00007	Penal Code	Section 12		section_body		12. 13.Rep	0	0	1	1	0	0			
SL00008	Penal Code	Section 14		section_body		14. by any	1	0	1	1	0	0			
SL00009	Penal Code	Section 15		section_body		15. The wo	0	0	1	0	0	0			
SL00010	Penal Code	Section 16		section_body		16. The wo	1	0	1	1	0	0			
SL00011	Penal Code	Section 17		section_bo	5	17. official	0	1	1	1	0	0			
SL00012	Penal Code	Section 17	illustration	illustration	5	Illustration	0	0	1	1	0	0			
SL00013	Penal Code	Section 18A		section_body		18A. The w	0	0	1	1	0	0			
SL00014	Penal Code	Section 19		section_bo	6	19. any of t	0	0	1	1	1	0			
SL00015	Penal Code	Section 19	illustration	illustration	6	Illustration	0	0	0	1	0	0			
SL00016	Penal Code	Section 19	explanatio	explanatio	6	Explanatio	0	0	1	1	0	0			
SL00017	Penal Code	Section 19	explanatio	explanatio	6	Explanatio	1	0	1	1	0	0			
SL00018	Penal Code	Section 20		section_bo	8	20. The wo	0	1	1	0	0	0			
SL00019	Penal Code	Section 20	1	subsection	8	(1) 'Wrongf	0	0	1	1	0	0			
SL00020	Penal Code	Section 20	2	subsection	8	(2) 'Wrongf	0	0	1	1	0	0			
SL00021	Penal Code	Section 20	3	subsection	8	(3) A perso	0	0	1	1	0	0			
SL00022	Penal Code	Section 20	4	subsection	8	(4) A perso	0	0	1	1	0	0			
SL00023	Penal Code	Section 22		section_body		22. Whoev	0	0	1	1	0	0			
SL00024	Penal Code	Section 23		section_body		23. A perso	0	1	1	1	0	0			
SL00025	Penal Code	Section 24		section_body		24. has suf	0	0	1	0	0	0			
SL00026	Penal Code	Section 25		section_body		25. When p	0	0	1	1	1	0			

Figure 5: Extracted statute text in Penal Code candidates dataset

5.5 Dataset construction

After extraction and review, the cleaned provisions were organised into a master legal simplification dataset. The dataset was designed to store not only the original and simplified text but also metadata that made it possible to trace each provision back to its statute, section, page range and legal unit type.

The main structure of the dataset is shown in Table 5.

Table 5: Output columns of master dataset

Field name	Description
id	Unique identifier for each dataset row
statute	Name of the statute, such as Maintenance Act or Penal Code
section_number	Main section number of the legal provision
subsection	Subsection number where applicable
heading	Heading or title connected to the provision
chapter	Chapter reference where applicable
part	Part reference within the statute
source_text	Original extracted legal provision
source_file	Source file from which the provision was extracted
page_start	Starting page of the provision in the source PDF
page_end	Ending page of the provision in the source PDF
unit_type	Type of extracted legal unit, such as section or subsection
include_status	Indicates whether the row is included or excluded after review
notes	Manual notes about extraction quality or review decisions
simplified_text	Plain-English version of the original legal provision

The most important part of the dataset is the relationship between source_text and simplified_text. The source_text field stores the original statutory provision, while the simplified_text field stores the plain-English version. The simplified version was written to make the legal text easier to understand while still preserving the original legal meaning.

The simplification process paid close attention to legal details that could not be changed. Numbers, dates, fines, ages and imprisonment terms were preserved. Conditions and exceptions were also preserved because removing them could change when the law applies. Legal roles such as Court, Magistrate, parent, spouse and child were kept as well, since these roles affect the meaning of the provision.

The simplified text was added through annotation batches. The full dataset was divided into smaller batches because completing the whole dataset at once would have been difficult to manage. Each batch was filled with simplified text and then merged back into the master dataset. This made the dataset easier to review, manage and correct.

example_i	act_name	section_id	unit_label	unit_type	page_start	source_tex	has_modal	has_excep	has_numb	has_roles	needs_cor	bad_fragm	simple_tex	split	notes	pilot_category
SL00023	Penal Code	Section 22		section_body		Whoever d	0	0	1	1	0	0	A person a	train		definition
SL00135	Penal Code	Section 10	explanatio	explanatio	30	A person w	0	0	1	1	0	0	A person ir	train		definition
SL00022	Penal Code	Section 20	4	subsectior	8	A person is	0	0	1	1	0	0	A person d	train		definition
SL00045	Penal Code	Section 35		section_bo	11	When an o	0	0	1	1	0	0	f a crime is	train		definition
SL00049	Penal Code	Section 37		section_body		A person is	0	0	1	1	0	0	A person a	train		definition
SL00073	Penal Code	Section 54		section_body		Sentence c	1	0	1	1	0	0	A pregnant	train		duty_punishment
SL00075	Penal Code	Section 56		section_body		In every ca	1	1	1	1	0	0	If someone	train		duty_punishment
SL00077	Penal Code	Section 57		section_body		No female	1	0	1	1	0	0	Women ca	train		duty_punishment
SL00160	Penal Code	Section 107		section_body		Whenever	1	1	1	1	0	0	If a person	train		duty_punishment
SL00166	Penal Code	Section 110		section_body		Whoever a	1	0	1	1	0	0	If a person	train		duty_punishment
SL00198	Penal Code	Section 135		section_bo	45	Whoever a	1	1	1	1	0	0	If a person	train		exception_proviso
SL00193	Penal Code	Section 129		section_body		Whoever a	1	1	1	1	0	0	If a person	train		exception_proviso
SL00168	Penal Code	Section 111		section_body		Whoever, i	1	1	1	0	0	0	If a person	train		exception_proviso
SL00120	Penal Code	Section 91	1	subsectior	26	There is no	1	1	1	1	0	0	If a public s	train		exception_proviso
SL00098	Penal Code	Section 78	provided_t	proviso	20	Provided th	1	1	0	1	0	0	If a crime r	dev		exception_proviso
SL00213	Penal Code	Section 149		section_body		Whoever a	1	0	1	1	0	0	If a person	dev		offence_definition
SL00177	Penal Code	Section 114		section_bo	40	Whoever w	1	0	1	0	0	0	If a person	dev		offence_definition
SL00184	Penal Code	Section 120		section_body		Whoever b	1	0	1	1	0	0	A person m	test		offence_definition
SL00187	Penal Code	Section 122		section_body		Whoever c	1	0	1	0	0	0	If a person	test		offence_definition
SL00202	Penal Code	Section 13	explanatio	explanatio	46	Explanatio	1	0	0	0	0	0	A gathering	test		offence_definition

Figure 6: Master legal simplification dataset containing source and simplified legal text.

id	statute	section_nu	source_tex	simplified_text
PC_S76	Penal Code	76	Nothing is	Nothing is an offence if it is done by a child who is above 12 years of age but under 14, and who had not yet reached enough maturity of understanding to appreciate the nature and consequences of their conduct on that o
PC_S79	Penal Code	79	In cases wl	Where an act is not an offence unless it is done with a particular knowledge or intent, a person who does that act while intoxicated is to be treated as if they had the same knowledge they would have had if they had not bee
PC_S80	Penal Code	80	Nothing, w	Nothing is an offence because of any harm it may cause â€" or that the doer intends or knows it is likely to cause â€" to a person who is above 18 years of age and who has given consent, whether express [clearly stated] c
PC_S81	Penal Code	81	Nothing, w	Nothing is an offence because of any harm it may cause, or that the doer intends or knows is likely to cause, to a person for whose benefit it is done in good faith and who has given consent â€" whether express or implied
PC_S82	Penal Code	82	Nothing, w	Nothing done in good faith for the benefit of a person under 12 years of age, or of a person of unsound mind, is an offence because of any harm it may cause â€" or that the doer intends or knows is likely to cause â€" to th
PC_S83	Penal Code	83	A consent	A consent is not a valid consent for the purposes of any section of this Code if: (a) it is given by a person under fear of injury, or under a mistaken belief about the facts, and the person doing the act knows or has reason to b
PC_S84	Penal Code	84	The except	The exceptions in sections 80, 81, and 82 do not apply to acts that are offences independently of any harm they may cause â€" or be intended or likely to cause â€" to the person giving or on whose behalf consent was give
PC_S85	Penal Code	85	Nothing is	Nothing is an offence because of any harm it may cause to a person for whose benefit it is done in good faith, even without that person's consent, if: the circumstances make it impossible for that person to give consent, o
PC_S86	Penal Code	86	No commu	No communication made in good faith is an offence because of any harm it causes to the person to whom it is made, if it is made for that person's benefit. Illustration: A, a surgeon, honestly tells a patient that the patient c
PC_S87	Penal Code	87	Except mu	Nothing is an offence if it is done by a person who is compelled to do it by threats that, at the time, give that person reasonable grounds to fear that they will immediately be killed if they do not comply. This exception does r
PC_S88	Penal Code	88	Nothing is	Nothing is an offence because it causes, or is intended or known to be likely to cause, any harm, if that harm is so minor that no reasonable person of ordinary sensibility would complain of it.
PC_S89	Penal Code	89	Nothing is	Nothing is an offence if it is done in the exercise of the right of private defence [self-defence or defence of others or property]. Every person has this right, subject to the restrictions â€"
PC_S91	Penal Code	91	When an a	Where an act would otherwise be a particular offence but is not that offence because of the youth, lack of maturity, unsoundness of mind, or intoxication of the person doing it, or because of a mistaken belief on that pers
PC_S92_1	Penal Code	92	There is no	There is no right of private defence against an act done â€" or attempted â€" by a public servant acting in good faith under the authority of their office, if that act does not give reasonable grounds to fear death or grievous h
PC_S92_2	Penal Code	92	There is no	There is no right of private defence against an act done â€" or attempted â€" on the direction of a public servant acting in good faith under the authority of their office, if that act does not give reasonable grounds to fear de
PC_S92_3	Penal Code	92	There is no	There is no right of private defence in situations where there is time to seek protection from the public authorities.
PC_S92_4	Penal Code	92	The right o	The right of private defence never extends to causing more harm than is necessary for the purpose of defence. Explanation 1: A person is not denied the right of private defence against an act done by a public servant unle
PC_S93	Penal Code	93	The right o	The right of private defence of the body extends, subject to the restrictions in section 92, to the deliberate causing of death or any other harm to the attacker, if the attack that gives rise to the right falls into any of the follow
PC_S94	Penal Code	94	If the offen	If the attack does not fall into any of the categories listed in section 93, the right of private defence of the body does not extend to deliberately causing the death of the attacker. It does extend, subject to the restrictions in s
PC_S95	Penal Code	95	The right o	The right of private defence of the body begins as soon as a reasonable fear of danger to the body arises from an attempt or threat to commit an offence, even if the offence has not yet been committed. It continues for as l
PC_S96	Penal Code	96	The right o	The right of private defence of property extends, subject to the restrictions in section 92, to deliberately causing death or any other harm to the wrongdoer, if the offence that gives rise to the right falls into any of the follow
PC_S97	Penal Code	97	If the offen	If the offence that gives rise to the right of private defence of property is theft, mischief, or criminal trespass not falling into any of the categories listed in section 96, the right does not extend to deliberately causing death.
PC_S98	Penal Code	98	Firstly- The	(1) The right of private defence of property begins when a reasonable fear of danger to the property begins. (2) The right of private defence of property against theft continues until the offender has escaped with the proper
PC_S99	Penal Code	99	If in the exe	If, while exercising the right of private defence against an attack that gives reasonable grounds to fear death, the defender is in a situation where they cannot effectively exercise that right without risking harm to an innoce
PC_S100	Penal Code	100	A person a	A person abets [encourages, assists, or conspires in] the doing of a thing if they: (1) instigate [urge or incite] any person to do that thing; or (2) take part in any conspiracy to do that thing; or (3) intentionally aid, by any act o

Figure 7: Example annotation batch used to complete simplified text.

5.6 Baseline simplification model evaluation

A baseline model evaluation stage was completed to compare different large language models before the simplification model was chosen for the system. The aim was to avoid assuming that one model was best without actually testing it. Each model was tested on the same legal provisions with the same simplification prompt so the results could be compared on a like-for-like basis.

The models tested are FLAN-T5-Large, Gemma 3 4B IT, Mistral 7B Instruct v0.3, Qwen3 8B and Llama 3.2 3B Instruct. These models are selected because they represent different instruction-following and transformer-based model options.

Separate notebooks were created for each model. Each notebook follows the same structure. It loads the dataset, loads the model, generates simplified output, calculates metrics, saves a clean output CSV and creates a summary table image. This makes the experiments more organised and easier to reproduce.

Each model's output was stored separately so the results would not get mixed together when they were compared later.

example_id	act_name	section_id	unit_type	split	source_text	simple_text	qwen3_output	qwen3_fkgl	qwen3_ari	qwen3_rougeL	qwen3_bertscore_f1
SL00023	Penal Code	Section 22	section_body	train	Whoever does anything with the intention of causing wrongful gain to one person, or wrongful loss to another person, is said to do that thing ' dishonestly '.	A person acts dishonestly if they intend to cause wrongful gain to one person or wrongful loss to another.	If someone does something with the intention of making another person lose unfairly, or causing unfair gain to themselves or someone else, then that action is considered 'dishonest'.	15.137142857142859	18.643214285714286	0.2127659574468085	0.9072620868682861
SL00135	Penal Code	Section 100	explanation	train	A person who. by willful misrepresentation or by willful concealment of a material fact which he is bound to disclose, voluntarily causes or procures, or attempts to cause or procure, a thing to be done, is said to instigate the doing of that thing.	A person instigates an act if they deliberately lie or hide an important fact they had a duty to disclose, and this causes or tries to cause another person to do that act.	If someone intentionally lies or hides a important fact that they are required to tell, and this leads to or tries to lead to a certain action being done, then that person is said to have instigated the action.	16.56358974358975	18.842307692307692	0.41666666666666663	0.9231606721878052
SL00022	Penal Code	Section 20	subsection	train	A person is said to lose wrongfully when such person	A person does not only lose property wrongfully	A person loses wrongfully when they are wrongfully	13.371818181818181	13.548181818181817	0.36363636363636365	0.9103165864944458

Figure 8: Generated simplified outputs of Qwen 3 8B.

5.7 Evaluation metrics used for baseline comparison

The baseline model outputs are evaluated using readability and semantic similarity metrics, since legal simplification has two goals at once. The output has to be easier to read, but it also has to preserve the meaning of the original provision.

FKGL and ARI are used as readability metrics. These scores help show whether the generated output is easier to read than the original legal text. Lower scores mean that the text is easier to understand.

ROUGE-L and BERTScore F1 are used to compare the model-generated simplification with the human-written reference simplification. ROUGE-L checks sequence overlap while BERTScore F1 gives a semantic similarity score. This is useful because a good simplification may use different wording from the reference but still preserve the meaning.

Table 6: Evaluation metrics used in baseline evaluation

Metric	**Purpose**
FKGL	Measures reading grade level
ARI	Estimates the education level needed to understand the text
ROUGE-L	Measures overlap with the reference simplification
BERTScore F1	Measures semantic similarity with the reference simplification

The metric results are saved for each model. These results are then used to build the model comparison sheet. The comparison helps identify which model gives the best balance between readability and meaning preservation.

Qwen3 8B - Baseline Evaluation Summary

Model	Rows	Avg FKGL	Avg ARI	Avg ROUGE-L	Avg BERTScore F1
Qwen3 8B	20	15.916	18.351	0.438	0.925

Figure 9: Summary table produced for the Qwen 3 8B model.

5.8 Model comparison and simplification model selection

After running the baseline notebooks, the model results are compared using the average metric values. Each model produces a clean output CSV, a metric summary table and values for the master comparison sheet. The automatic comparison includes FLAN-T5-Large, Gemma 3 4B IT, Mistral 7B Instruct v0.3, Qwen3 8B and Llama 3.2 3B Instruct.

The baseline evaluation produces closely matched results across the strongest candidate models. Qwen3 8B, Gemma 3 4B IT and Mistral 7B Instruct v0.3 perform strongly across the main metrics. Qwen3 8B records the highest BERTScore F1 score of 0.925. This suggests that its generated simplifications have the strongest semantic similarity to the human-written reference simplifications among the models tested. Its readability scores are also competitive, with an average FKGL of 15.916 and an average ARI of 18.351. These values place Qwen3 8B close to the lowest readability scores achieved by Gemma 3 4B IT.

However, the automatic scores are not treated as complete proof of legal faithfulness. BERTScore F1 only shows semantic similarity with the reference simplification. It does not prove that the model preserves every legal condition, exception, punishment or legal role correctly. Because of this, a qualitative output review is also carried out before selecting the final simplification model.

Model	Rows	Avg FKGL	Avg ARI	Avg ROUGE-L	Avg BERTScore F1
FLAN-T5-Large	20	22.484	26.009	0.333	0.897
Gemma 3 4B IT	20	15.793	17.689	0.439	0.924
Llama 3.2 3B Instruct	20	17.101	19.412	0.381	0.906
Mistral 7B Instruct v0.3	20	17.002	19.157	0.455	0.924
Qwen3 8B	20	15.916	18.351	0.438	0.925

Green = best in metric Red = worst in metric

Figure 10: Baseline model comparison sheet used to select the simplification model.

The qualitative review focused mainly on Qwen3 8B and Gemma 3 4B IT because these two models were the strongest candidates after the automatic comparison. Their generated outputs were compared against the original source provisions and the human-written simplified references. The review focused on whether the models preserved numbers, legal roles, conditions, exceptions and the overall legal meaning of the provision. One example from the qualitative review showed why readability scores alone could not be used to select the final model. Gemma 3 4B IT omitted the statutory citation to section 282 of the Code of Criminal Procedure Act No. 15 of 1979 from one simplified output. This citation was important because it helped define the scope of the protection given by the provision and allowed the reader to trace the rule back to its source legislation.

Gemma 3 4B IT also changed the phrase “shall not be pronounced” into “may not order”. This was treated as a legal meaning issue because “shall not” expresses an absolute prohibition, while “may not” can sound more discretionary to a lay reader. In the same example, Qwen3 8B preserved both the statutory citation and the prohibitory wording. Although Qwen3 8B produced a slightly less readable output, it was more accurate for this provision because it kept the legal reference and the legal force of the original text.

This example supported the final model decision. Gemma 3 4B IT was useful for producing readable text, but Qwen3 8B provided a better balance between readability, semantic similarity and legal faithfulness in the reviewed outputs.

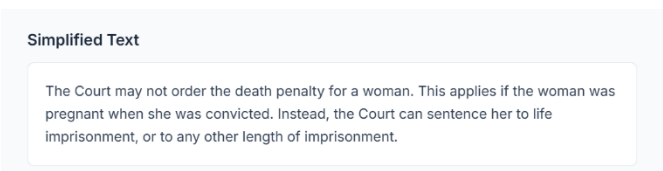

Figure 11: Output of Gemma 3 4B IT

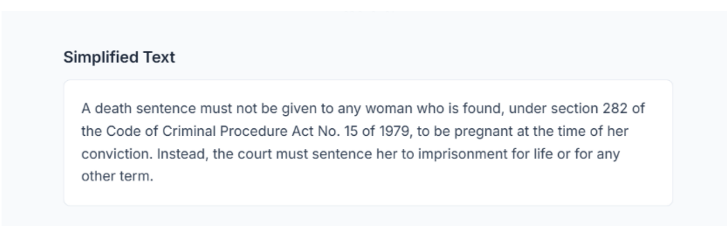

Figure 12: Output of Qwen 3 8B

5.9 Frontend prototype

A frontend prototype was developed for ClearClause using HTML, Tailwind CSS and JavaScript. Tailwind CSS was used because it made the interface easier to style without requiring a large amount of custom CSS.

The frontend included a landing section, simplification page, reusable navbar, legal disclaimer banner, text area for entering a legal provision, simplify button, loading animation and result display area. It also included sections for the simplified output, atomic claims, source spans and backend errors

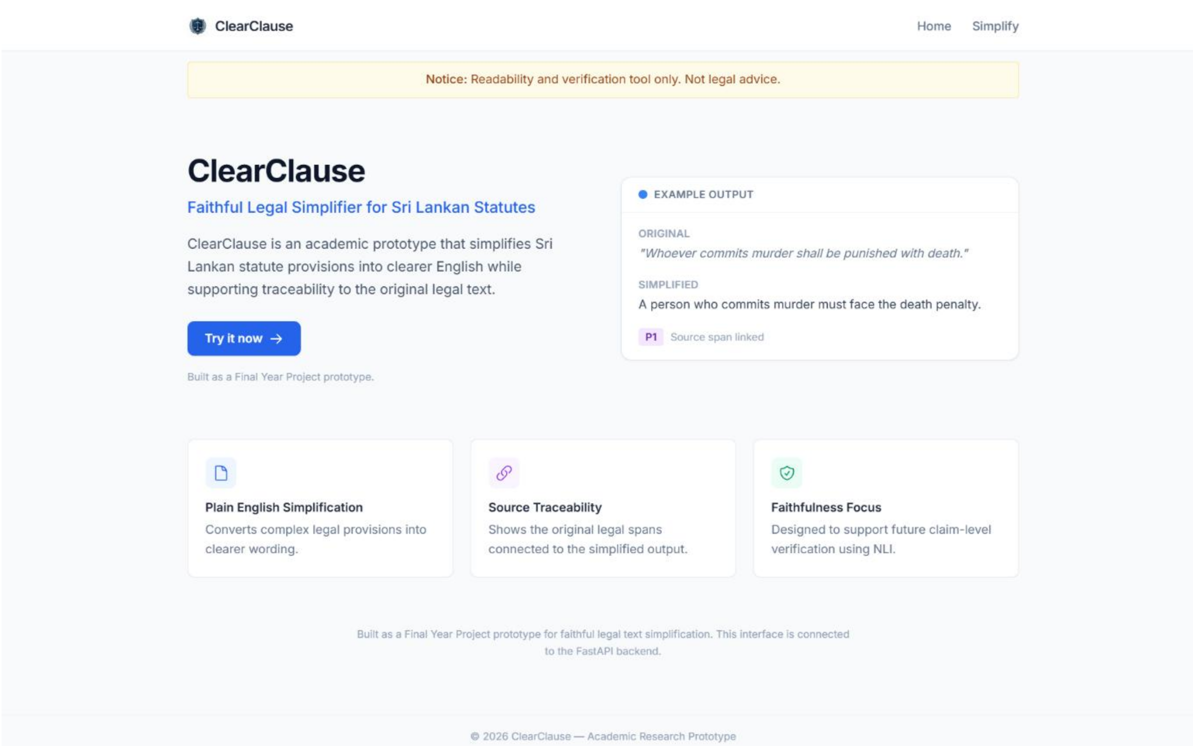

Figure 13: Landing page of ClearClause

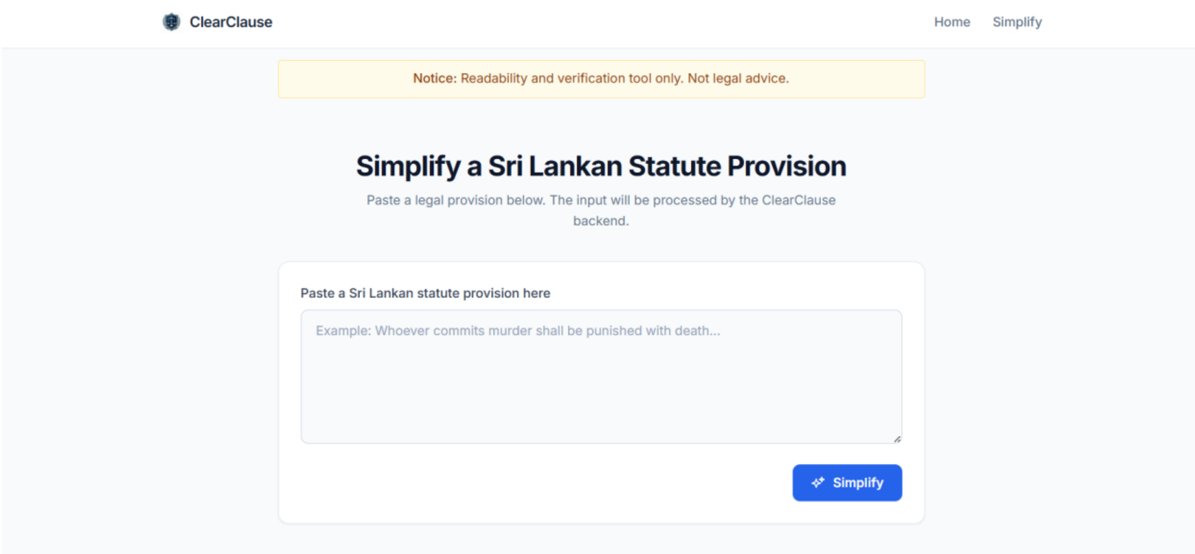

Figure 14: Simplification interface with legal input area.

5.10 Frontend and backend integration

The frontend was connected to the FastAPI backend through the /pipeline endpoint. This changed the frontend from a static prototype into an interface that could send user input to the backend and display the returned result.

When a user entered a legal provision and clicked the simplify button, the frontend sent a POST request to the backend. (POST /pipeline)

The request body contained the legal text entered by the user.

The returned data was displayed in the result section. The simplified legal text appeared first, followed by the extracted atomic claims and the source spans. This made the system output easier to inspect because the user could view both the simplified version and the source evidence structure together.

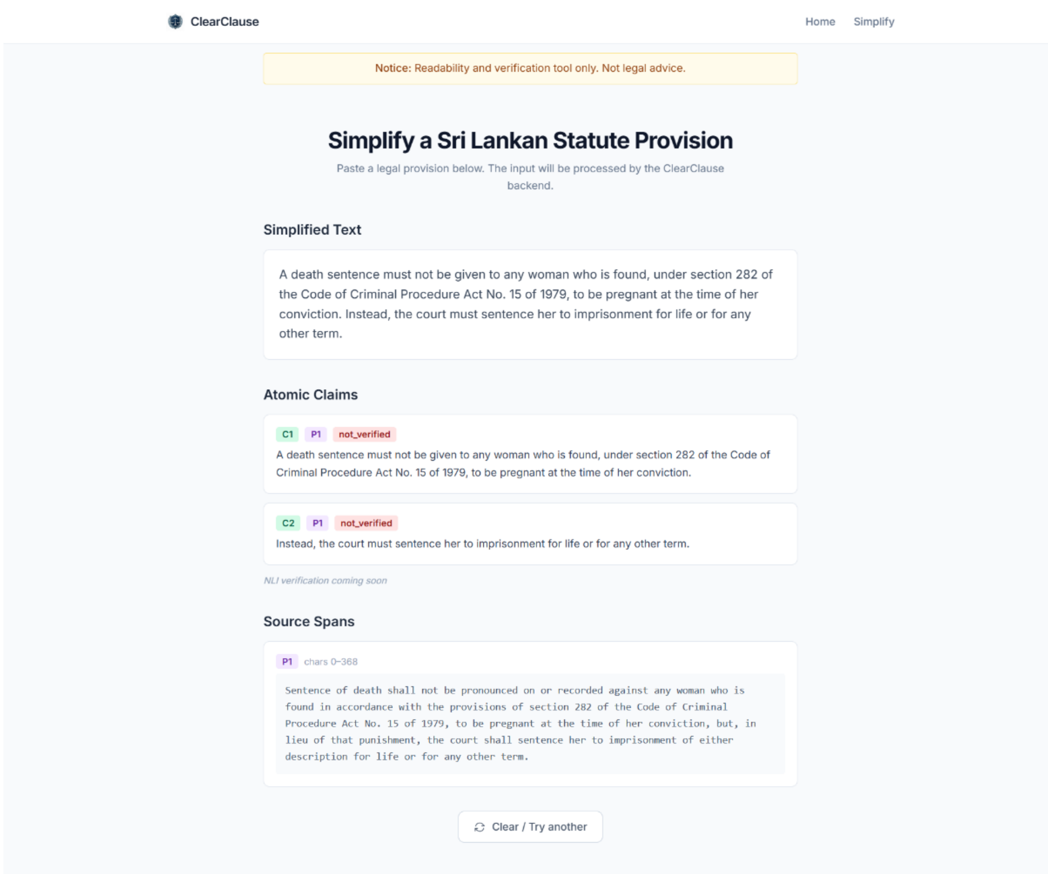

Figure 15: Frontend displaying simplified text, claims and source spans returned from the backend.

5.11 FastAPI backend pipeline

The backend was developed using FastAPI. It was organised into separate pipeline stages because the system had more than one task. The backend needed to build spans, call the simplification model, extract claims and link claims to source evidence. Separating these stages made the system easier to test and debug.

The main endpoint is POST /pipeline

This endpoint runs the full pipeline. It receives the legal text from the frontend and returns the spans, simplified text and claims.

The backend also includes debugging endpoints. These are useful because each part of the pipeline can be tested separately through FastAPI's /docs page.

The /health endpoint checks whether the backend is running. The /spans endpoint tests the deterministic span builder. The /simplify endpoint tests the Qwen 3 8B simplification call. The /claims endpoint tests atomic claim extraction. The /evidence endpoint tests evidence linking.

This structure provides better debugging access because errors can be isolated to one stage instead of checking the full pipeline every time.

5.12 Deterministic span builder

A deterministic span builder was implemented to divide the pasted legal provision into source spans. These spans were labelled as P1, P2, P3 and so on. Each span also stored the span text, start offset and end offset.

The span builder used rule-based splitting instead of machine learning. This made the process more transparent because the source text was divided using known legal markers and sentence boundaries. This approach was useful for legal text because statutes often contain conditions, provisos, explanations and illustrations that need to remain traceable.

Each span contains the following details.

Table 7: Details in each span

Field	Description
span_id	Span label such as P1, P2 or P3
text	Text contained in the span
start_offset	Starting character position
end_offset	Ending character position

The span builder detected legal markers such as “Provided that”, “Explanation”, “Illustration”, subsection markers and legal connectors such as “Unless”, “Except” and “Notwithstanding”. These markers helped split the provision in a way that kept important legal parts visible.

This stage was important because the later verification process depended on traceability. Each simplified claim needed to be linked back to the original statutory text. The P1 to Pn span structure gave the system a clear evidence base.

5.13 Qwen 3 simplification and claim extraction

Qwen 3 8B was used in the current backend design for two generation tasks. The first task was legal simplification. The original source text was sent to Qwen 3 8B, and the model returned a simplified plain-English version.

The simplification prompt asked the model to simplify the provision without adding new legal facts or removing important legal details. The output was expected to preserve obligations, punishments, conditions, exceptions, numbers and legal roles.

The second task was atomic claim extraction. In this stage, the simplified text was sent to Qwen 3 again, and the model returned claims in a structured JSON format.

Each claim was expected to contain one legal fact. This was important because a full simplified paragraph may contain several legal statements. If the whole paragraph is checked as one block, it becomes difficult to identify which exact part is unsupported. Claim extraction makes the verification process more detailed.

This claim-level structure prepared the system for DeBERTa verification because each claim could later be used as the hypothesis in an NLI pair.

Figure 16: Simplified legal text generated using Qwen 3 8B.

Figure 17: Atomic claims extracted from the simplified text.

5.14 Prompts used in the system

Prompt design was an important part of the ClearClause system because the simplification output depended heavily on how the model was instructed. Since the project dealt with legal text, the prompt could not simply ask the model to "make the text simple". It also needed to remind the model to preserve legal meaning, conditions, exceptions, punishments, numbers and legal roles. Three prompts were used in the project. The first type was used in the backend pipeline for the actual ClearClause prototype. The second type was used during baseline evaluation so that different models could be tested under similar conditions

5.14.1 Backend simplification prompt

The backend simplification prompt was used when a user entered a legal provision into the ClearClause frontend. It was designed to keep the output faithful to the original legal text. The system needed a simplified version that was easier to read, but still legally accurate. For this reason, the prompt directly instructed the model not to add new legal information or remove important legal details.

The prompt is given below.

```
SIMPLIFY_SYSTEM = """You are a faithful legal simplification assistant for Sri Lankan statutes."""

SIMPLIFY_USER = """Task:
Rewrite the legal provision in plain English for a non-lawyer.

Rules:
- Preserve the exact legal meaning.
- Do not summarise.
- Do not remove any rule, condition, exception, punishment or legal consequence.
- Keep all numbers, dates, ages, fines, time periods and imprisonment terms exactly the same.
- Preserve legal force:
  - shall means must
  - may means can or is allowed to
  - shall not means must not
- Preserve legal roles such as Court, Magistrate, police officer, parent, spouse, child, guardian and employer.
- Use short sentences.
- Use simple words.
- Do not add examples.
```

```
- Do not give legal advice.
- Do not mention that you are an AI.

Return only the simplified text. No headings. No bullet points unless the original has multiple separate rules.

Legal provision:
{source_text}"""
```

5.14.2 Backend atomic claim extraction prompt

After the simplified text is generated, a second prompt is used to extract atomic claims. This step prepares the output for evidence linking and later NLI verification. Each claim is expected to contain only one legal fact.

```
CLAIM_SYSTEM = """You are a legal claim extraction assistant.

Extract atomic but complete legal claims from simplified legal text.

Output ONLY a valid JSON array. No explanation, no preamble, no markdown fences.

Format:
[
  {{"claim_id": "C1", "claim_text": "..."}},
  {{"claim_id": "C2", "claim_text": "..."}}
]"""

CLAIM_USER = """Task:
Break the simplified legal text into atomic claims.

Atomic claim rules:
- Each claim must contain only one legal fact.
- Do not combine multiple rules into one claim.
- Do not add anything not stated in the simplified text.
- Preserve legal force: must, can, must not.
- Preserve all numbers, dates, ages, fines, punishments, time periods and legal roles.
- If a condition applies, include the condition in the claim.
- If an exception applies, include the exception in the claim.
- Do not explain the law.
- Do not give legal advice.

Return JSON only in this exact format:
[
  {{"claim_id": "C1", "claim_text": "..."}},
  {{"claim_id": "C2", "claim_text": "..."}}
]

Simplified legal text:
{simplified_text}"""
```

5.14.3 Baseline evaluation prompt

During baseline evaluation, the same general simplification task was given to each candidate model. This made the comparison fairer because every model received the same instruction style and the same legal provisions. The baseline evaluation prompt was slightly simpler than the backend prompt because its main purpose was to compare model outputs, rather than to run the full frontend-backend pipeline.

```
 {"role": "system", "content": "You are a careful legal simplification assistant."},
        {"role": "user", "content": f"""Simplify the Sri Lankan legal provision below into plain English.

Rules:
- Preserve the legal meaning as closely as possible.
- Keep SHALL, MAY, SHALL NOT.
- Keep numbers, dates, fines, punishments, conditions, exceptions and legal roles.
- Do not add examples, commentary or legal advice.

Provision:
{source_text}

Plain English:"""}
```

5.15 Evidence linking

A basic evidence linking method was implemented using keyword overlap. This was not the final NLI verification stage. Instead, it was an early evidence-selection method that connected each atomic claim to the most relevant source span.

The method tokenised the claim and each source span. Stopwords were removed so that common words did not influence the comparison too much. After that, the system compared the remaining keywords and selected the source span with the highest overlap.

This method was simple, but it worked well for the midpoint version because it provided a clear first connection between generated claims and original legal text. It also made the output more explainable because each claim had an evidence span instead of appearing without any source reference.

The evidence linking stage was important for the planned NLI verification step. In the final verifier, the selected source span could be used as the premise, while the atomic claim could be used as the hypothesis.


```json
  {
    "span_id": "P1",
    "text": "No female shall in any case be punished with whipping.",
    "start": 0,
    "end": 54
  },
  {
    "span_id": "P2",
    "text": "Nor shall any person who may be sentenced to death or to imprisonment for more than five years be punished with whipping.",
    "start": 55,
    "end": 176
  }
],
"claims": [
  {
    "claim_id": "C1",
    "claim_text": "No woman must be punished with whipping."
  },
  {
    "claim_id": "C2",
    "claim_text": "A person sentenced to death or imprisonment for more than five years must not be punished with whipping."
  }
]
```


Figure 18: Atomic claims linked to source evidence spans

5.16 NLI verification design

The DeBERTa-based NLI verification stage was the next major part of the system. It was not fully integrated at the midpoint stage, but the system was already structured for it. The span builder, claim extraction and evidence linking stages all prepared the input needed for NLI verification.

The verifier classifies the relationship using three labels.

Table 8: NLI labels used in ClearClause verification

Label	**Meaning in ClearClause**
Entailment	The claim is supported by the source span
Contradiction	The claim conflicts with the source span
Neutral	The claim is not clearly supported by the source span

The planned verifier uses microsoft/deberta-v3-small. The NLI dataset is created from the completed simplification dataset by using legal provisions as premises, atomic claims as hypotheses and entailment, contradiction or neutral as labels.

At the midpoint stage, this remained the main technical task for the next phase. The existing pipeline already prepared the source spans and claims, so the verifier could be connected after the NLI dataset was created and the model was trained or fine-tuned.

5.17 Current system status

By the midpoint stage, the project had completed the main foundation needed for the final prototype. The legal data sources were selected, the PDF extraction process was completed, extraction issues were reviewed and the master simplification dataset was created. The annotation batches were also completed and merged back into the dataset.

The baseline model evaluation was completed for the selected candidate models. Qwen 3 8B was selected as the current simplification model based on the comparison results. The frontend prototype was implemented using HTML, Tailwind CSS and JavaScript. Frontend-backend integration had started through the /pipeline endpoint. The FastAPI backend included the main pipeline design and debugging endpoints. The deterministic span builder and keyword-based evidence linking method were also implemented.

The main remaining work was the final NLI stage. This included final dataset cleaning, creating the NLI dataset, training or fine-tuning the DeBERTa verifier and connecting the verifier output to the backend pipeline.

5.18 Chapter summary

The implementation and data collection work completed for ClearClause up to the midpoint stage has been presented in this chapter. The chapter explained the technology stack, legal data collection process, PDF extraction workflow and construction of the master legal simplification dataset using the Penal Code and the Maintenance Act. It also described the baseline model evaluation process, where candidate models were compared using FKGL, ARI, ROUGE-L and BERTScore F1 before

Qwen3 8B was selected as the simplification model based on metric results and qualitative comparison with Gemma 3 4B IT.

The frontend prototype, FastAPI backend, deterministic span builder, backend prompts, atomic claim extraction and keyword-based evidence linking were also explained. The current implementation shows that the main foundation of the ClearClause pipeline has been developed, while the remaining work focuses mainly on creating the NLI dataset and integrating the DeBERTa-based verifier.

REFERENCES

Akter, M., Çano, E., Weber, E., Dobler, D. and Habernal, I. (2025) 'A comprehensive survey on legal summarization: challenges and future directions', arXiv preprint arXiv:2501.17830. Available at: https://arxiv.org/abs/2501.17830 (Accessed: 10 January 2026).

Ariai, F., Mackenzie, J. and Demartini, G. (2025) 'Natural language processing for the legal domain: a survey of tasks, datasets, models and challenges', ACM Computing Surveys, 58(6), article 163, pp. 1–37. doi: 10.1145/3777009.

BCS, The Chartered Institute for IT (2025) BCS Code of Conduct for members. Available at: https://www.bcs.org/membership-and-registrations/become-a-member/bcs-code-of-conduct/ (Accessed: 7 June 2026).

Cemri, M., Çukur, T. and Koç, A. (2022) 'Unsupervised simplification of legal texts', arXiv preprint arXiv:2209.00557. Available at: https://arxiv.org/abs/2209.00557 (Accessed: 10 January 2026).

Choi, J., Honda, U., Watanabe, T. and Inui, K. (2023) 'Explainable natural language inference in the legal domain via text generation', Transactions of the Japanese Society for Artificial Intelligence, 38(3), pp. C-MB6_1–C-MB6_11. doi: 10.1527/tjsai.38-3_C-MB6.

Curran, D., Sporne, V., Frermann, L. and Paterson, J. (2025) 'Place matters: comparing LLM hallucination rates for place-based legal queries', arXiv preprint arXiv:2511.06700. Available at: https://arxiv.org/abs/2511.06700 (Accessed: 10 January 2026).

Dahl, M., Magesh, V., Suzgun, M. and Ho, D.E. (2024) ‘Large legal fictions: profiling legal hallucinations in large language models’, Journal of Legal Analysis, 16(1), pp. 64–93. doi: 10.1093/jla/laae003.

Data Protection Authority of Sri Lanka (2024) Guidelines and publications under the Personal Data Protection Act No. 9 of 2022. Colombo: Data Protection Authority of Sri Lanka. Available at: https://www.dpa.gov.lk/guidelines.php (Accessed: 7 June 2026).

Dehghani, F., Dehghani, R., Naderzadeh Ardebili, Y. and Rahnamayan, S. (2025) ‘Large Language Models in Legal Systems: A Survey’, Humanities and Social Sciences Communications, 12, article 1977. doi: 10.1057/s41599-025-05924-3.

Fernando, W.S., Handapangoda, M., Samindi, N., Gamage, M. and Alwis, D. (2024) ‘LawKey: Law Constitution Chatbot’, in Proceedings of the 2024 8th SLAAI International Conference on Artificial Intelligence (SLAAI-ICAI). doi: 10.1109/SLAAI-ICAI63667.2024.10844946.

Garimella, A., Sancheti, A., Aggarwal, V., Ganesh, A., Chhaya, N. and Kambhatla, N. (2022) ‘Text simplification for legal domain: insights and challenges’, in Proceedings of the Natural Legal Language Processing Workshop 2022. Abu Dhabi: Association for Computational Linguistics, pp. 296–304. doi: 10.18653/v1/2022.nllp-1.28.

Huang, L., Yu, W., Ma, W., Zhong, W., Feng, Z., Wang, H., Chen, Q., Peng, W., Feng, X., Qin, B. and Liu, T. (2025) ‘A survey on hallucination in large language models: principles, taxonomy, challenges and open questions’, ACM Transactions on Information Systems, 43(2), article 42, pp. 1–55. doi: 10.1145/3703155.

ICO (2023) Guidance on AI and data protection. Wilmslow: Information Commissioner's Office. Available at: https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/artificial-intelligence/guidance-on-ai-and-data-protection/ (Accessed: 7 June 2026).

Jayawardena, L., Wiratunga, N., Abeyratne, R., Martin, K., Nkisi-Orji, I. and Weerasinghe, R. (2024) 'SCaLe-QA: Sri Lankan case law embeddings for legal QA', in CEUR Workshop Proceedings: SICSARE-LLM Workshop 2024, pp. 47–55. Available at: https://ceur-ws.org/Vol-3822/short6.pdf (Accessed: 10 January 2026).

Koreeda, Y. and Manning, C.D. (2021) 'ContractNLI: a dataset for document-level natural language inference for contracts', in Findings of the Association for Computational Linguistics: EMNLP 2021. Punta Cana: Association for Computational Linguistics, pp. 1907–1919. Available at: https://aclanthology.org/2021.findings-emnlp.164/ (Accessed: 10 January 2026).

Laban, P., Schnabel, T., Bennett, P.N. and Hearst, M.A. (2022) 'SummaC: re-visiting NLI-based models for inconsistency detection in summarization', Transactions of the Association for Computational Linguistics, 10, pp. 163–177. doi: 10.1162/tacl_a_00453.

Magesh, V., Surani, F., Dahl, M., Suzgun, M., Manning, C.D. and Ho, D.E. (2025) 'Hallucination-free? assessing the reliability of leading AI legal research tools', Journal of Empirical Legal Studies, 22(2), pp. 216–245. doi: 10.1111/jels.12413.

Maynez, J., Narayan, S., Bohnet, B. and McDonald, R. (2020) 'On faithfulness and factuality in abstractive summarization', in Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics. Association for Computational Linguistics, pp. 1906–1919. Available at: https://aclanthology.org/2020.acl-main.173/ (Accessed: 7 June 2026).

Muralidharan, A. (2021) Automatic English Text Simplification for Statutes. Master's thesis. München: Technische Universität München. Available at: https://www.cs.cit.tum.de/fileadmin/w00cfj/sebis/thesis/211015_Muralidharan_Thesis.pdf (Accessed: 10 January 2026).

Parliament of Sri Lanka (2022) Personal Data Protection Act, No. 9 of 2022. Colombo: Parliament of Sri Lanka. Available at: https://www.parliament.lk/uploads/acts/gbills/english/6242.pdf (Accessed: 7 June 2026).

Saunders, M.N.K., Lewis, P. and Thornhill, A. (2019) Research Methods for Business Students. 8th edn. Harlow: Pearson.

Scirè, A., Ghonim, K. and Navigli, R. (2024) 'FENICE: Factuality Evaluation of summarization based on Natural Language Inference and Claim Extraction', in Findings of the Association for Computational Linguistics: ACL 2024. Bangkok: Association for Computational Linguistics, pp. 14148–14161. doi: 10.18653/v1/2024.findings-acl.841.

Sharma, A., Singh, A.A., Singh, A., Shrivastava, D. and Kumar, P. (2025) 'Hybrid legal language simplification for Indian judiciary: leveraging LegalBERT and summarization techniques', in Proceedings of the 2025 International Conference on Next Generation Information System Engineering (NGISE). IEEE. doi: 10.1109/NGISE64126.2025.11085325.

Tang, X., Li, J., Hu, K., Nan, D., Li, X., Zhang, X., Sun, W. and Xie, S. (2025) 'CogniBench: a legal-inspired framework and dataset for assessing cognitive faithfulness of large language models', arXiv preprint arXiv:2505.20767. Available at: https://arxiv.org/abs/2505.20767 (Accessed: 10 January 2026).

Tonmoy, S.M.T.I., Zaman, S.M.M., Jain, V., Rani, A., Rawte, V., Chadha, A. and Das, A. (2024) ‘A comprehensive survey of hallucination mitigation techniques in large language models’, arXiv preprint arXiv:2401.01313. Available at: https://arxiv.org/abs/2401.01313 (Accessed: 10 January 2026).

UNESCO (2021) Recommendation on the Ethics of Artificial Intelligence. Paris: United Nations Educational, Scientific and Cultural Organization. Available at: https://unesdoc.unesco.org/ark:/48223/pf0000380455 (Accessed: 7 June 2026).